

LEPL1109 Statistics & Data science: part I.

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Roadmap

Semaine	Dates		Cours Magistral	Hackathons	TP
1	14-09-2026	au	18-09-2026	Stat 1	X
2	21-09-2026	au	25-09-2026	Stat 2	X
3	28-09-2026	au	02-10-2026	Stat 3	X
4	05-10-2026	au	09-10-2026	Stat 4	X (H1)
5	12-10-2026	au	16-10-2026	Stat 5	X
6	19-10-2026	au	23-10-2026	Q&A H1	X
7	26-10-2026	au	30-10-2026	suspension des cours	
8	02-11-2026	au	06-11-2026	Data 1	X (TP1)
9	09-11-2026	au	13-11-2026	Data 2	X (TP2)
10	16-11-2026	au	20-11-2026	Data 3	X (TP3)
11	23-11-2026	au	27-11-2026	Data 4	
12	30-11-2026	au	04-12-2026	X (H2) - 2 séances	
13	07-12-2026	au	11-12-2026	X (H2) - 1 séance	
14	14-12-2026	au	18-12-2026	Data 5	X (drill)

Roadmap

	Monday	Tuesday	Wednesday	Thursday	Friday
8:30 - 10:30		Serie 1	Serie 3	Serie 6	Serie 9
10:45 - 12:45			Serie 4	Serie 7	
14:00 - 16:00	Lecture		Serie 5	Serie 8	Serie 10
16:15 - 18:15		Serie 2	Serie 11		

TP's and Hackathons: REGISTER to a group on Moodle!

Hackathons

- ▶ Hackathons are small pedagogical group works in Python (Jupyter notebook to download).
- ▶ 2 activities of this type are planned during the semester : 1 in statistic and 1 in data.
- ▶ The sessions of exercises during these weeks are Q&A about hackathons
- ▶ Hackathons are done by groups of 5 to 6 students

Back to probabilities

This chapter reviews some concepts of probability, used later in statistics.

- ▶ If we consider an experiment, the set of possible results ω of this experiment is noted Ω
- ▶ We associate to each event A of the sample space Ω a measure of probability, $P(A)$, such that $P(A) \in [0, 1]$ and $P(\Omega) = 1$.

A **Random variable** is a measurable function from the sample space Ω to the set of real numbers $\mathbb{R}$, $X : \Omega \rightarrow \mathbb{R}$. We denote it by a capital letter and its realization by a lowercase letter ($X(\omega) = x$).

Random variable

- ▶ The set of all possible values of a r.v. X is the state space (or the **range** of X) of the r.v. : $\{x \in \mathbb{R} \mid x = X(\omega), \omega \in \Omega\}$.
- ▶ A r.v. is called **discrete** if its state space has a finite or countable number of elements.
- ▶ A r.v. is called **continuous** if it takes arbitrary real values between a minimum and a maximum. The state space is either an interval of $\mathbb{R}$, or $\mathbb{R}$.
- ▶ **Example:** In the context of a chemical experiment, let consider the r.v. X = time of reaction in ms (one thousandth of a second) after application of a stimulant. In the normal conditions, this time varies between 263 ms and 613 ms. So $P(263 \leq X \leq 613) = 1$ and $\text{Range}(X) = [263; 613]$.

Probability distribution

The probability mass function (pmf) $p(x)$ of a discrete random variable X is a function that associates to all values x of X a probability $P(X = x)$:

$$p(x) = P(X = x).$$

If the range of X is $R = \{x_1, x_2, \dots\}$ then $p(x) > 0$ if $x \in R$ and $p(x) = 0$ if $x \notin R$. Furthermore $\sum_i p(x_i) = 1$.

Once the pmf is known, we calculate the probability that $P(X \in I)$ where I is a discrete subset of $\mathbb{R}$ as

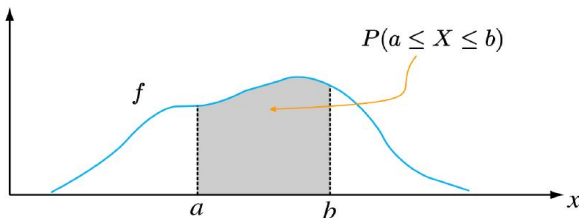
$$P(X \in I) = \sum_{i: x_i \in I} p(x_i)$$

Probability density function (pdf)

If the random variable is **continuous**, we cannot enumerate mass probabilities for all possible values of x .

Let X be a continuous random variable. The **probability density function (pdf)** of X is the function $f(x) \geq 0$ such that for any $I \subset \mathbb{R}$, we have that

$$P(X \in I) = \int_I f(x) dx$$



Cumulative distribution function (cdf)

The **cumulative distribution function (cdf)**, $F(x)$, of a random variable (discrete or continuous) X indicates for each possible value of x the probability that X takes the value equal or less than x .

$$F(x) = P(X \leq x)$$

In the discrete case, if the state space (range) of $X = \{x_1, x_2, \dots\}$ then

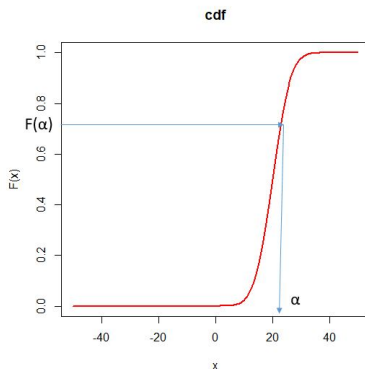
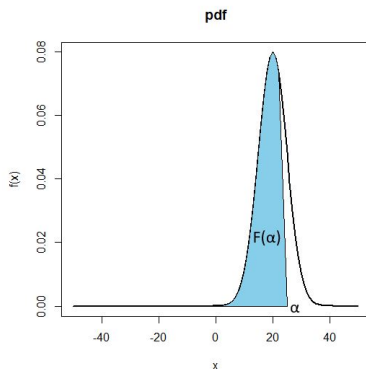
$$F(x) = \sum_{i: x_i \leq x} p(x_i)$$

In the continuous case,

$$F(x) = \int_{-\infty}^x f(u) du$$

Link between the pdf & cdf

The pdf is the derivative of the cdf: $f(x) = \frac{d}{dx}F(x)$.



Furthermore, given that the integral of the pdf $f(x)$ is a probability, we must have $\int_{-\infty}^{\infty} f(x)dx = P(X \in \text{range}(X)) = 1$.

Mathematical expectation

The expectation of a r.v. X is noted $\mathbb{E}(X) = \mu_X$. If X is a discrete r.v.

$$\mu_X = \sum_{x \in \text{range}(X)} x p(x).$$

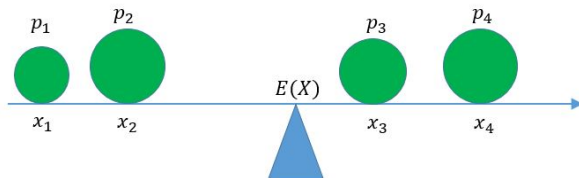
If X is a continuous r.v. with a density $f(x)$, the expectation of X is equal to

$$\mu_X = \int_{-\infty}^{+\infty} x f(x) dx$$

We will see later that if we observe n outcomes of a random variable x_i , the empirical mean $\bar{x} = \frac{\sum_{i=1:n} x_i}{n}$ converges to the expected value μ_X (see “Law of Large Numbers”) Numpy: `np.mean(X)`

Mathematical expectation

- ▶ If X is a discrete r.v., its expectation μ_X is the center of gravity of points



- ▶ **Property:** Let X and Y be two random variables then

$$\mathbb{E}(X + Y) = \mathbb{E}(X) + \mathbb{E}(Y)$$

$$\mathbb{E}(aX + b) = a\mathbb{E}(X) + b$$

Mathematical expectation

The expected value of a real-valued function $h(\cdot)$ of a discrete r.v. X is equal to

$$\mathbb{E}(h(X)) = \sum_{x \in \text{range}(X)} h(x) p(x).$$

If X is a continuous r.v. with a density $f(x)$, the expectation of $h(X)$ is equal to

$$\mathbb{E}(h(X)) = \int_{-\infty}^{+\infty} h(x) f(x) dx$$

Important remark: in general we do not have (check it on a example)

$$\mathbb{E}(h(X)) \neq h(\mathbb{E}(x))$$

$$\mathbb{E}(XY) \neq \mathbb{E}(X)\mathbb{E}(Y)$$

Notice that the k^{th} moment of X is defined by $\mathbb{E}(X^k)$.

Variance

The variance of a random variable X is denoted by

$$\text{Var}(X) = \mathbb{V}(X) = \sigma_X^2$$

(discrete or continuous) and is defined as

$$\sigma_X^2 = \mathbb{E} \left((X - \mu_X)^2 \right)$$

The **standard deviation** is the square root of the variance $\sigma_X = \sqrt{\mathbb{V}(X)}$.

The standard deviation is a measure of dispersion of the probability function around its balancing point (the mean). When the variance is close to zero, outcomes of X are concentrated around the mean. On the contrary, if the variance is big, realizations of X may be very far from each other.

Properties of the variance. If X and Y are two r.v. (discrete or continuous) and $a, b \in \mathbb{R}$, then

- ▶ $\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2$
- ▶ $\text{Var}(a) = 0$
- ▶ $\text{Var}(a + bX) = b^2 \text{Var}(X)$
- ▶ In general, $\text{Var}(X + Y) \neq \text{Var}(X) + \text{Var}(Y)$ except if X and Y are independent

- ▶ The proofs are left to the reader and do not present any difficulty.
- ▶ If we observe n outcomes of a random variable x_i , the empirical variance $s^2 = \frac{\sum_{i=1:n} (x_i - \bar{x})^2}{n-1}$ is used as estimator of σ_X^2 (details provided later) Numpy: `np.std(X)`

Law of Large Numbers (LLN)

Law of Large Numbers: Let $X_{i=1,\dots,n}$ be a sequence of uncorrelated r.v.'s with the same expectation $\mu_X = \mathbb{E}(X_i)$ and variance σ_X^2 . When $n \rightarrow \infty$, the sample mean $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ converges in probability to μ_X :

$$\forall \epsilon > 0 \quad \lim_{n \rightarrow \infty} P(|\bar{X}_n - \mu_X| \geq \epsilon) = 0.$$

We denote this by $\bar{X}_n \xrightarrow{P} \mu_X$.

Formal proof: Expectations/Variances are additive, and $\mathbb{V}(aY) = a^2\mathbb{V}(Y)$. Then $\mathbb{E}(\bar{X}_n) = \mu_X$ and

$$\mathbb{V}(\bar{X}_n) = \frac{1}{n^2} \mathbb{V}\left(\sum_{i=1}^n X_i\right) = \frac{\sigma_X^2}{n}$$

The variance tends to zero when $n \rightarrow \infty$.

Quantiles of a distribution

Quantile of a distribution Let p be a probability between 0 and 1 and X be a r.v;. The number q_p satisfying the relation

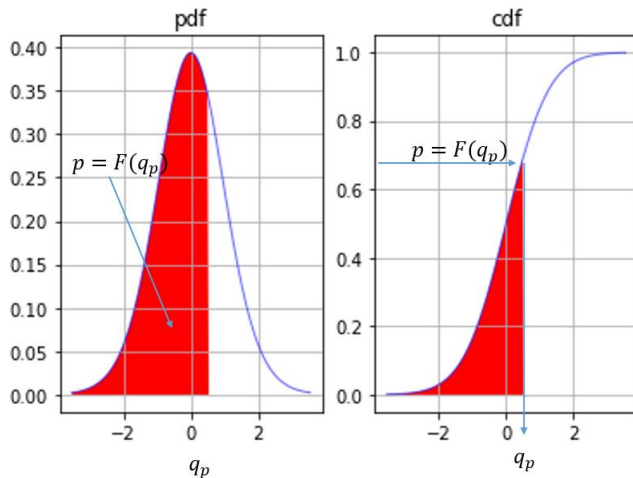
$$P(X \leq q_p) = p$$

is the quantile of order p for X . If X is continuous and if its cdf $F(x)$ is invertible then $q_p = F^{-1}(p)$.

If $p = 5\%$ then X is below $q_{5\%}$ with a probability of 5% and X is above $q_{5\%}$ with a probability of 95%. The quantile is used in the banking/insurance industry as measure of risk (Value At Risk).

Quantiles of a distribution

Python scipy.stats: `sc.norm.ppf(0.01, loc= $\mu$ , scale= $\sigma$ )`



Quantiles of a distribution

Example

- ▶ Let the r.v. X be the operating life (in days) until failure of a certain device.
- ▶ Assume that the density of X is

$$f(x) = \frac{1}{50} e^{-\frac{x}{50}} I_{[0, \infty)}(x)$$

- ▶ To calculate the quantile of order p of X , we must solve the following equation for q

$$p = \frac{1}{50} \int_0^q e^{-\frac{x}{50}} dx$$

- ▶ The solution is given by

$$q_p = -50 \ln(1 - p)$$

- ▶ Thus, in 5% of case the device operating time will be smaller than 2.56 days!

Linear transformation. Let X be a continuous r.v. with density $f_X(x)$. The density (pdf) of $Y = a + bX$ is given by

$$f_Y(y) = \frac{1}{|b|} f_X\left(\frac{y-a}{b}\right)$$

Function of random variables

Proof. Assume that $b < 0$ By definition of Y ,

$$\begin{aligned}P(Y \leq y) &= P(a + bX \leq y) \\&= P(X \geq \frac{y - a}{b}) \quad \text{because } b < 0 \\&= \int_{\frac{y - a}{b}}^{+\infty} f_X(x) dx\end{aligned}$$

We derive w.r.t. y and obtaine

$$\begin{aligned}f_Y(y) &= \frac{\partial}{\partial y} P(Y \leq y) \\&= -\frac{1}{b} f_X\left(\frac{y - a}{b}\right),\end{aligned}$$

and we conclude as $b < 0$, **end.** In python, `scipy.stats: loc = a`
and scale = b

Descriptive statistics and first data visualization tools

Introduction

Some vocabulary:

- ▶ **Population:** collection of all possible outcomes, responses, measurements, or counts that are of interest.
- ▶ **Sample:** a subset of population.
- ▶ **Descriptive statistics:** branch of statistics that involves the organization, summarization and display of data.

Download files “DescriptiveStatistics.py” and “DurationData.csv”.
Run the Python script and make sure that the path is well the one of the directory containing the csv file.

Example: supply chain

- ▶ We consider an automotive manufacturing chain with a succession of workstations:



Some parts of motors are bolted together by a team of workers. We measure the time spent by team of 3 and 5 workers on 1000 motors (sample size $n = 1000$).

Introduction

- ▶ We have the following table of durations in minutes:

Duration		
Motor #	3 Workers, X	5 Workers, Y
1	24.23	15.23
$\vdots$	$\vdots$	$\vdots$
1000	18.5	8.56

- ▶ The time spent by teams of 3 and 5 workers are random quantities respectively denoted by X and Y .
- ▶ Each observation in the sample is a realization of this random variable. The i^{th} realizations are denoted by x_i and y_i for $i = 1, \dots, 1000$. These records are stored in the file "DurationData.csv".

Mean, Median, Quantiles

Which descriptive statistics could we use to analyze this dataset? We naturally think to the mean and the median in order to give a central value for the data set.

- ▶ The **SAMPLE MEAN** is the average value for a set of n observations. This is the center of gravity of data.

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

- ▶ The **SAMPLE MEDIAN** is the value $q_{0.5}$ such that 50% of realizations are below $q_{0.5}$ and 50% are above $q_{0.5}$. The sample median is also called a quantile of order 0.5.

Mean, Median, Quantiles

The concept of median can be extended by the notion of sample quantile. For a given $p \in (0, 1)$, the p^{th} sample quantile, noted q_p , is a value such that a proportion p of the sample is smaller than q_p and a proportion $1 - p$ of observations is larger than q_p .

► **SAMPLE QUANTILE** : Given a set of realizations $x_1, \dots, x_n$ we define quantiles as follows:

► We sort the value by ascending order:

$$x_{(1)} \leq \dots \leq x_{(n)},$$

these values are called the order statistics of the original sample.

► q_p is the order statistics

$$q_p = x_{(1+(n-1)p)}$$

Mean, Median, Quantiles

Example: supply chain ,

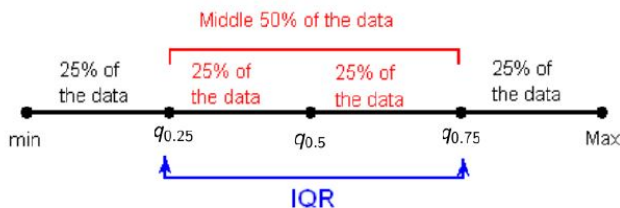
```
In [22]: data = pd.read_csv("durationData.csv", sep=';')
...: stat = data.describe()
...: print(stat)
...:
```

	workers3	workers5
count	1000.000000	1000.000000
mean	19.897557	9.893154
std	5.080540	6.336737
min	2.371155	0.384120
25%	16.512532	5.388633
50%	19.799504	8.560589
75%	23.121497	13.201465
max	37.203430	43.190108

We import the data stored in a “csv” file in a dataframe (library “pandas”, `read_csv()`) and we use the command `describe()`. Other command, numpy : `np.mean()`, `np.std()`, `np.corrcoef()`, `np.quantile(x,q)`

RANGE, IQR, OUTLIERS

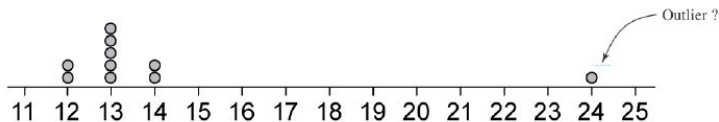
- ▶ The **RANGE** is the difference between the maximum and the minimum: $x_{max} - x_{min}$
- ▶ The **INTER-QUARTILE RANGE (IQR)** is the difference between symmetric quartiles, e.g. $IQR = q_{0.75} - q_{0.25}$



Python command: library "scipy.stat": `iqr()`

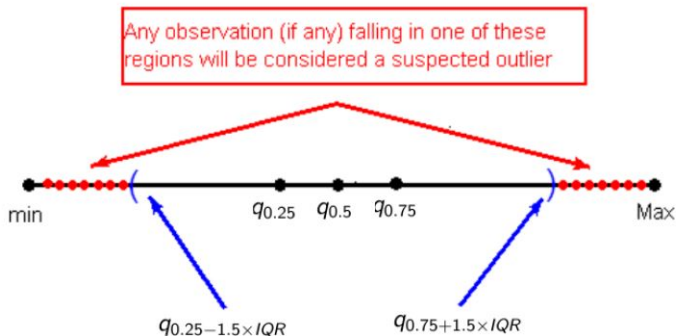
RANGE, IQR, OUTLIERS

- ▶ **OUTLIERS** : observations that appear to be far away from the rest of the data set. OUTLIERS may be due to errors of measurement or of encoding.
- ▶ In this case, it should be corrected or removed from the sample.



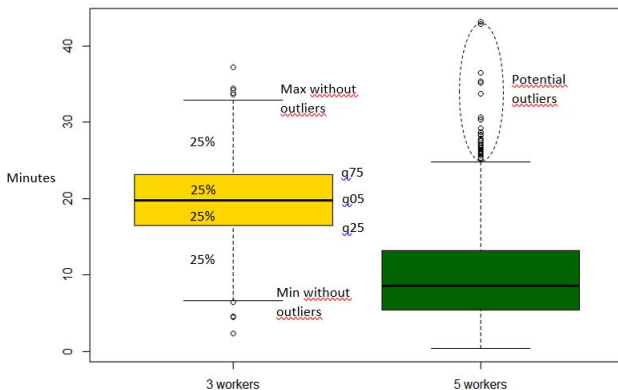
RANGE, IQR, OUTLIERS

An observation is considered as a suspected outlier if the value is below $q_{0.25} - 1.5 \times IQR$ or above $q_{0.75} + 1.5 \times IQR$



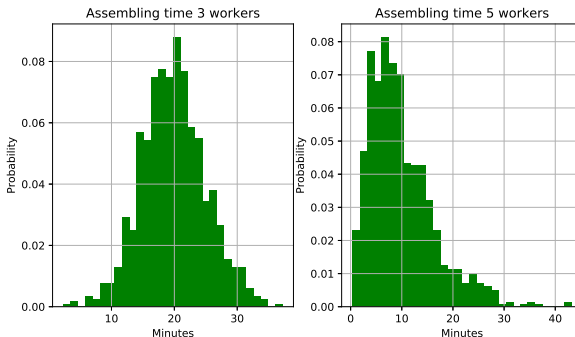
Graphical analysis

- ▶ The **BOXPLOT** graphically represents the distribution of observations by reporting the following statistics: min, $q_{0.25}$, $q_{0.50}$, $q_{0.75}$ and max. It also reports suspected outliers (IQR criterion).



Graphical analysis

- ▶ The **HISTOGRAM** : This tool aims to visualize the distribution of numerical observations. An histogram breaks the range of values into intervals and counts the proportion of observations in each bin. **This is the empirical pdf or pmf.**



Python command, library “matplotlib.pyplot”: `hist()`, `subplot()`

Variance and standard deviation

- ▶ The mean provides a central value for the data set. How do we measure the dispersion/spread around this average?
- ▶ Solution, the **sample VARIANCE**. We sum up the quadratic spread between observations and the mean:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

- ▶ the **sample STANDARD DEVIATION** is the square root of the variance: $s = \sqrt{s^2}$ (Numpy command: `std( ,ddof=1)` , `mean()`, `median()`)

Duration		
	3 Workers, X	5 Workers, Y
mean	19.89	9.89
median	19.79	8.56
Std. Dev.	5.08	6.34

Independence & linear dependence

Independent random variables

Two r.v. X and Y are **independent** ($X \perp\!\!\!\perp Y$) if for every A and $B \in \mathbb{R}$, we have

$$P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$$

Consequence of independence:

- ▶ $p(x, y) = p_X(x)p_Y(y)$ when X and Y are discrete
- ▶ $f(x, y) = f_X(x)f_Y(y)$ when X and Y are continuous

Example of independent r.v.:

- ▶ X : variation of the BEL 20 stocks index,
- ▶ Y : temperatures in Louvain-La-Neuve

Covariance

When two random variables are not independent, how can we measure the degree of dependence between them? The linear dependence is measured by the covariance.

Let X and Y be random variables. The **covariance** between X and Y is

$$\begin{aligned}\sigma_{XY} &= \text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] \\ &= \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y) \\ &= \mu_{XY} - \mu_X \mu_Y\end{aligned}$$

The covariance can be thought of as the mean of matches and mismatches among the pair (X, Y) . The covariance is positive when the matches outweigh the mismatches and is negative when the mismatches outweigh matches.

Covariance

- ▶ If we observe n outcomes $(x_k)_{k=1,\dots,n}$ and $(y_k)_{k=1,\dots,n}$ of r.v.'s X and Y . The empirical covariance is estimated as follows

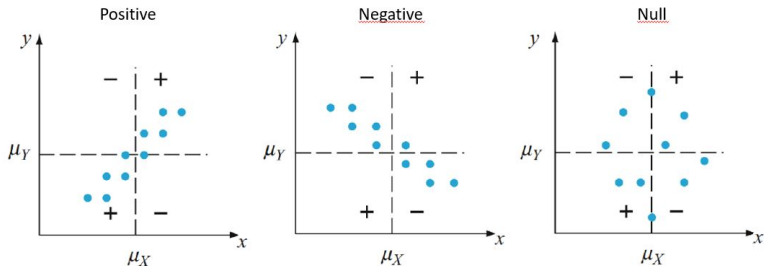
$$\sigma_{XY} \approx s_{XY} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

- ▶ Covariance command in Python : `numpy, np.cov(m , rowvar=True)`

Covariance

Three types of dependence:

- ▶ $\mathbb{C}(X, Y) > 0$: X and Y move in the same direction
- ▶ $\mathbb{C}(X, Y) < 0$: X and Y move in opposite directions



Covariance

From the definition of covariance, we can deduce the following properties

- ▶ $\mathbb{C}(X, Y) = \mathbb{C}(Y, X)$
- ▶ $\mathbb{C}(X, X) = \mathbb{V}(X)$ and $\mathbb{C}(a, X) = 0$ for all $a \in \mathbb{R}$
- ▶ if $X \perp\!\!\!\perp Y$ then $\mathbb{C}(X, Y) = 0$ (X and Y are uncorrelated)
- ▶ $\mathbb{C}(aX + bY, Z) = a\mathbb{C}(X, Z) + b\mathbb{C}(Y, Z)$
- ▶ $\mathbb{V}(aX + bY) = a^2\mathbb{V}(X) + b^2\mathbb{V}(Y) + 2ab\mathbb{C}(X, Y)$

Attention if $\mathbb{C}(X, Y) = 0 \not\Rightarrow X \perp\!\!\!\perp Y$!!! Example, $Y = X^2$ with $\mathbb{C}(X, Y) = 0$:

		X		
		-1	0	1
Y	0	0	1/3	0
	1	1/3	0	1/3

Correlation

What type of dependence does the correlation measure? The covariance is a measure of **LINEAR** dependence between two r.v.

To explain this, we define a scaled measure of dependence.

Reason? The size of the covariance depends upon the scale of variables $\mathbb{C}(aX, Y) = a\mathbb{C}(X, Y)$.

Let X and Y be random variables. The **correlation** between X and Y is defined as

$$\begin{aligned}\rho_{XY} &= \frac{\mathbb{C}(X, Y)}{\sqrt{\mathbb{V}(X)\mathbb{V}(Y)}} \\ &= \frac{\sigma_{XY}}{\sigma_X\sigma_Y}\end{aligned}$$

Correlation

According the Cauchy Schwarz inequality, we have that $|\sigma_{XY}| \leq \sigma_X \sigma_Y$ with equality if and only if $Y = aX + b$. Then the correlation has the following properties:

- ▶ ρ_{XY} is scale invariant
- ▶ $-1 \leq \rho_{XY} \leq 1$
- ▶ If $\rho_{XY} = 1$ then $Y = a + bX$ with $b \in \mathbb{R}^+$ (Y proportional to X)
- ▶ If $\rho_{XY} = -1$ then $Y = a + bX$ with $b \in \mathbb{R}^-$ (Y inversely proportional to X)

The correlation is a measure of the linear dependence between two r.v. easier to interpret than σ_{XY} because it is “unit-free” and in $[-1, 1]$.

- ▶ If we observe n outcomes $(x_1, \dots, x_n)$ and $(y_1, \dots, y_n)$ of r.v. X and Y . The correlation is estimated by the empirical correlation:

$$\rho_{XY} \approx r_{XY} = \frac{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2}}$$

- ▶ Python command: `numpy, np.corrcoef(x,y)`.

Normal random variable and Central Limit Theorem

Normal distribution

A r.v. X follows a **Normal distribution** if its density is given by the following function

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right) \quad x \in \mathbb{R}$$

where $\mu \in \mathbb{R}$ and $\sigma \in \mathbb{R}^+$ are parameters. We note $X \sim \mathcal{N}(\mu, \sigma^2)$.

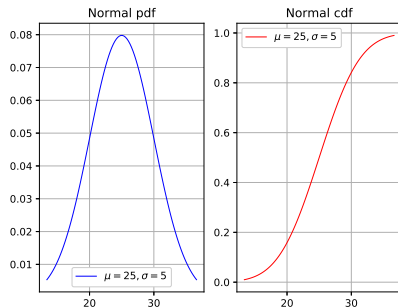
This is used for modeling e.g. the noise in signal processing. The parameters are easy to interpret:

$$\mathbb{E}(X) = \mu \quad , \quad \mathbb{V}(X) = \sigma^2$$

$$\underbrace{m_X(t)}_{(*)} := \mathbb{E}\left(e^{tX}\right) = \exp\left(\mu t + \frac{1}{2}t^2\sigma^2\right)$$

(*) moment generating function $\left.\frac{\partial^n m_X(t)}{\partial t^n}\right|_{t=0} = \mathbb{E}(X^n)$

Normal distribution



See file

statisticalDistributionPlot.py.

Python: `scipy.stats`: command

`norm.pdf(x, loc= $\mu$ , scale= $\sigma$ )`,

`norm.cdf(x, loc= $\mu$ , scale= $\sigma$ )` , and

quantiles with

`norm.ppf(x, loc= $\mu$ , scale= $\sigma$ )`

Standardization

If $X \sim \mathcal{N}(\mu, \sigma^2)$ then for any a and $b \neq 0$ then $a + bX$ is a normal r.v.

$$\mathcal{N}(a + b\mu, b^2\sigma^2)$$

If $X \sim \mathcal{N}(\mu_x, \sigma_x^2)$ and $Y \sim \mathcal{N}(\mu_y, \sigma_y^2)$ and covariance is $\sigma_{x,y} = \mathbb{C}(X, Y)$

$$aX + bY \sim \mathcal{N}(a\mu_x + b\mu_y, a^2\sigma_x^2 + b^2\sigma_y^2 + 2ab\sigma_{x,y})$$

If $X \sim \mathcal{N}(\mu, \sigma^2)$ then $\frac{X-\mu}{\sigma} \sim Z = \mathcal{N}(0, 1)$ (**standardization**) and

$$P(X \leq x) = P\left(Z \leq \frac{x-\mu}{\sigma}\right)$$

Central Limit Theorem (CLT)

Central Limit Theorem. Let $X_1, \dots, X_n$ be a sequence of iid (*) r.v.'s with $\mathbb{E}(X_i) = \mu$ and $\mathbb{V}(X_i) = \sigma^2$. As $n \rightarrow \infty$

$$Z_n = \sqrt{n} \frac{\bar{X}_n - \mu}{\sigma} \xrightarrow{d} Z \sim \mathcal{N}(0, 1)$$

i.e.

$$P(Z_n \leq z) \xrightarrow{n \rightarrow \infty} P(Z \leq z) \quad \forall z$$

Interpretation: for large n , whatever the distribution of X_i , the distribution of the sample mean $\bar{X}_n$ and of the sum $S_n = \sum_{i=1}^n X_i$ may be approached by a Normal distribution

$$\begin{aligned}\bar{X}_n &\sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right) \\ S_n &\sim \mathcal{N}(n\mu, n\sigma^2)\end{aligned}$$

(*) iid: independent identically distributed

Central Limit Theorem (CLT)

Example 2: The CLT may be used to approach the binomial distribution $Bi(n, p)$. Let $Y_n \sim Bi(n, p)$. Y_n is a sum of n independent Bernoulli r.v. X_i ($X_i = 1$ with probability p and zero otherwise):

$$Y_n = X_1 + \dots + X_n.$$

We have that $\mathbb{E}(X_i) = p$ and $\mathbb{V}(X_i) = p(1-p)$. Let $\hat{p}_n = \frac{Y_n}{n}$. By the CLT we have that

$$\sqrt{n} \frac{\hat{p}_n - p}{\sqrt{p(1-p)}} = \frac{Y_n - np}{\sqrt{np(1-p)}} \xrightarrow{d} \mathcal{N}(0, 1)$$

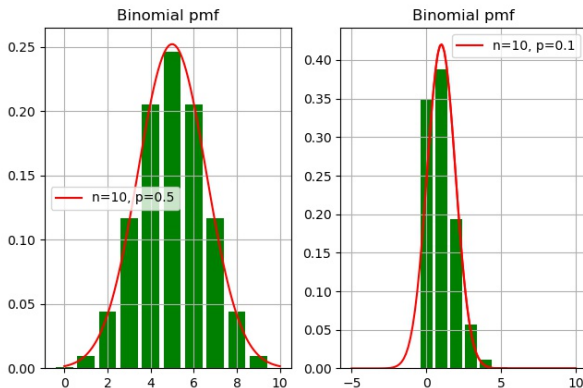
Equivalent formulation:

$$Bi(n, p) \approx \mathcal{N}(np, np(1-p))$$

This approximation will be rather good if p (and $1-p$) is not too small and n sufficiently large : $n > 9 \frac{\max(p, 1-p)}{\min(p, 1-p)}$.

Central Limit Theorem (CLT)

Approximation of a Binomial law $Bi(n, p)$ by a $\mathcal{N}(np, n(1 - p))$
(see Python file [TCLillustration.py](#)) :



Other fundamental random variables

Chi-square distribution

- ▶ The **chi-square** (χ^2) distribution with n degrees of freedom is the distribution of **a sum of the squares of n independent standard normal $\mathcal{N}(0, 1)$ r.v.**

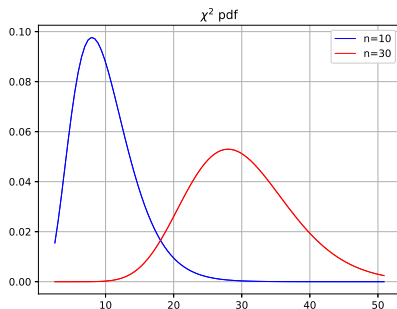
A r.v. X defined on $\mathbb{R}^+$ follows a χ^2 -**distribution** of parameters n , when its density (pdf) is given by

$$\frac{1}{2^{n/2}\Gamma(n/2)} x^{n/2-1} e^{-x/2}$$

where $\Gamma(\alpha) = \int_0^\infty t^{\alpha-1} e^{-t} dt$ (Gamma function).

- ▶ Then $\mathbb{E}(X) = n$ and $\mathbb{V}(X) = 2n$.
- ▶ The relation between the χ^2 and normal r.v.'s will play an important role in the second part of the course (statistical inference: hypothesis testing).

Chi-square distribution



See file

statisticalDistributionPlot.py.

Python: `scipy.stats`: command

`chi2.pdf(x,df=n)`,

`chi2.cdf(x,df=n)` , and quantiles

with `chi2.ppf(x,df=n)`

We will conclude this section by introducing two distributions playing an important role in statistical inference.

T distribution. Let Z and Y be two independent r.v. $Z \sim \mathcal{N}(0, 1)$ and $Y \sim \chi_n^2$ then

$$T_n = \frac{Z}{\sqrt{Y/n}}$$

is the **Student'T** r.v. with n degrees of freedom. Its pdf is

$$f_{T_n}(t) = \frac{\Gamma\left(\frac{n+1}{2}\right)}{\sqrt{n\pi}\Gamma\left(\frac{n}{2}\right)} \left(1 + \frac{t^2}{n}\right)^{-\frac{n+1}{2}} \quad t \in \mathbb{R}$$

F distribution. Let X and Y be two independent χ^2 r.v. with n_1 and n_2 degrees of freedom. Then

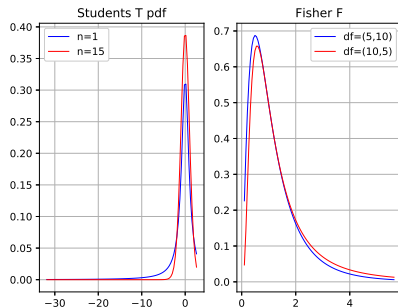
$$F_{n_1, n_2} = \frac{X/n_1}{Y/n_2}$$

is distributed as a **Fisher-Snedecor distribution** r.v. $\mathcal{F}(n_1, n_2)$ with n_1 and n_2 degrees of freedom. Its pdf is

$$f_{F_{n_1, n_2}}(z) = \frac{\Gamma\left(\frac{n_1+n_2}{2}\right) \left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}} z^{\left(\frac{n_1}{2}-1\right)}}{\Gamma\left(\frac{n_1}{2}\right) \Gamma\left(\frac{n_2}{2}\right) \left(1 + \frac{n_1}{n_2}z\right)^{\frac{n_1+n_2}{2}}} \quad t \in \mathbb{R}$$

Properties: if $F_{n_1, n_2} \sim \mathcal{F}(n_1, n_2)$ then $1/F_{n_1, n_2} \sim \mathcal{F}(n_2, n_1)$.

Student's T & Fischer S.



See file

statisticalDistributionPlot.py.

Python: `scipy.stats`: command

`t.pdf(x,df=n)`,

`t.cdf(x,df=n)` ,

`f.pdf(x,dfn= $n_1$ , dfd= $n_2$ )` ,

`f.cdf(x,dfn= $n_1$ , dfd= $n_2$ )`

Multivariate random variables

Introduction

A **random vector** $\mathbf{X} = (X_1, \dots, X_d)^\top$ is an application from $\Omega \rightarrow \mathbb{R}^p$, such that each coordinate is a random variable.

If $X_1, \dots, X_d$ are continuous r.v., the **multivariate probability distribution function** (pdf) of $\mathbf{X}$, $f(\cdot) : \mathbb{R}^p \rightarrow \mathbb{R}^+$ is such that for $\mathbf{x} = (x_1, \dots, x_d)^\top$:

$$\begin{aligned} P(\mathbf{X} \leq \mathbf{x}) &= P(X_1 \leq x_1, \dots, X_d \leq x_d) \\ &= \int_{-\infty}^{x_d} \dots \int_{-\infty}^{x_1} f(u_1, \dots, u_d) du_1 \dots du_d. \end{aligned}$$

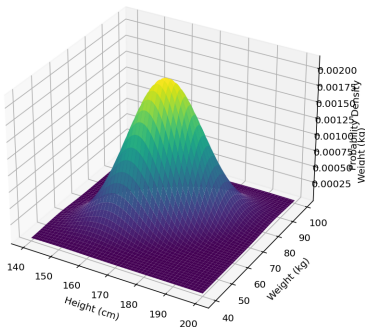
If $X_1, \dots, X_d$ are discrete r.v., its **multivariate probability mass function** (pmf), $p(\cdot) : \mathbb{R}^p \rightarrow [0, 1]$ is such that for $\mathbf{x} = (x_1, \dots, x_d)^\top$:

$$\begin{aligned} P(\mathbf{X} = \mathbf{x}) &= P(X_1 = x_1, \dots, X_d = x_d) \\ &= p(x_1, \dots, x_d). \end{aligned}$$

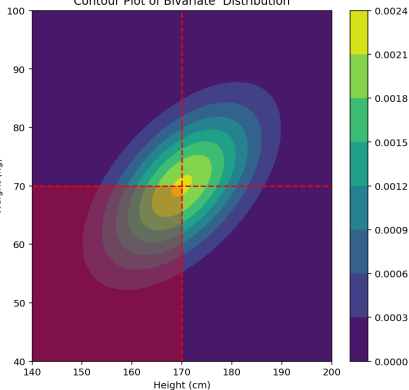
Introduction

Example 1: Human height X_1 and weight X_2 form a bivariate random variable. The integral of $f(x_1, x_2)$ over the shaded area on the right plot corresponds to $P(X_1 \leq 70\text{kg}, X_2 \leq 170\text{cm})$.

3D Surface Plot of Bivariate Distribution



Contour Plot of Bivariate Distribution



Example 2: Suppose we categorize student grades in two subjects, math and physics, into three levels: Low (0), Medium (1) and High (2).

Let X_1 and X_2 be the grades in math and physics:

		X_2		
$P(x_1, x_2)$		0	1	2
X_1	0	$p_{00} = 0.20$	$p_{01} = 0.05$	$p_{02} = 0.00$
	1	$p_{10} = 0.05$	$p_{11} = 0.25$	$p_{12} = 0.05$
	2	$p_{20} = 0.00$	$p_{21} = 0.05$	$p_{22} = 0.35$

We have that $\sum_{i=0}^{i=2} \sum_{j=0}^{j=2} p_{i,j} = 1$.

Marginal distributions

Let $\mathbf{X} = (X_1, \dots, X_d)^\top$ be a continuous random vector. The **marginal pdf** of X_k for $k = 1, \dots, d$ is

$$\begin{aligned} P(X_k \leq x) &= P(X_1 \leq \infty, \dots, X_1 \leq x, \dots, X_d \leq \infty) \\ &= \int_{-\infty}^{+\infty} \dots \int_{-\infty}^x \dots \int_{-\infty}^{+\infty} f(u_1, \dots, u_d) du_1 \dots du_d. \end{aligned}$$

The marginal pdf of X_k is a function $f_k : \mathbb{R} \rightarrow \mathbb{R}^+$ such that

$$\begin{aligned} f_{X_k}(x) &= \frac{\partial}{\partial x} P(X_k \leq x) \\ &= \int_{-\infty}^{+\infty} \dots \int_{-\infty}^{+\infty} f(u_1, \dots, u_{k-1}, x, u_{k+1}, \dots, u_d) du_1 \dots du_d. \end{aligned}$$

The bivariate distribution of X_i and X_j is

$$f_{X_i, X_j}(x_i, x_j) = \int_{-\infty}^{+\infty} \dots \int_{-\infty}^{+\infty} f(u_1, \dots, x_i, \dots, x_j, \dots, u_d) du_1 \dots du_d$$

Marginal distributions

Let $\mathbf{X} = (X_1, \dots, X_d)^\top$ be a discrete random vector. The **marginal probability mass function (pmf)** of X_k for $k = 1, \dots, d$ is

$$\begin{aligned} P(X_k = x_{k,j}) &= \sum_{i_d=1}^{m_d} \dots \sum_{i_{k+1}=1}^{m_{k+1}} \sum_{i_{k-1}=1}^{m_{k-1}} \dots \sum_{i_1=1}^{m_1} p(x_{1,i_1}, \dots, x_{k,j}, \dots, x_{1,i_d}) \\ &= p_{\bullet, \dots, j, \dots, \bullet} \end{aligned}$$

		X_2			
$P(x_1, x_2)$		0	1	2	
X_1	0	$p_{00} = 0.20$	$p_{01} = 0.05$	$p_{02} = 0.00$	$p_{0,\bullet} = 0.25$
	1	$p_{10} = 0.05$	$p_{11} = 0.25$	$p_{12} = 0.05$	$p_{1,\bullet} = 0.35$
	2	$p_{20} = 0.00$	$p_{21} = 0.05$	$p_{22} = 0.35$	$p_{2,\bullet} = 0.40$
		$p_{\bullet,0} = 0.25$	$p_{\bullet,1} = 0.35$	$p_{\bullet,2} = 0.40$	

Conditional distributions

Let us recall that for a small Δ , the pdf may be interpreted as a probability

$$f(x_1, \dots, x_d) \Delta^d \approx P(x_1 \leq X_1 \leq x_1 + \Delta, \dots, x_d \leq X_d \leq x_d + \Delta).$$

On the other hand the Bayes theorem states that for any subsets A and B of $\mathbb{R}^p$, the probability that $X \in A$, conditionally to $X \in B$ is:

$$P(\mathbf{X} \in A | \mathbf{X} \in B) = \frac{P(\mathbf{X} \in A \cap B)}{P(\mathbf{X} \in B)}$$

Then, if $d = 2$, we define the conditional pdf of the r.v. $X_1|X_2$ as follows:

$$f_{X_1|X_2}(x_1|x_2) = \frac{f(x_1, x_2)}{f_{X_2}(x_2)}.$$

By definition, after multiplying by Δ , we have:

$$f_{X_1|X_2}(x_1|x_2) \Delta \approx P(x_1 \leq X_1 \leq x_1 + \Delta | X_2 = x_2)$$

Conditional distributions

Let $\mathbf{X} = (X_1, \dots, X_d)^\top$ be a continuous or discrete random vector. We partition the vector into two disjoint sub-vectors $\mathbf{X}_A$ and $\mathbf{X}_B$. The **conditional pdf or pmf** of $\mathbf{X}_A | \mathbf{X}_B$ is

$$f_{\mathbf{X}_A | \mathbf{X}_B}(\mathbf{x}_A | \mathbf{x}_B) = \frac{f((\mathbf{x}_A, \mathbf{x}_B))}{f_{\mathbf{X}_B}(\mathbf{x}_B)},$$
$$p_{\mathbf{X}_A | \mathbf{X}_B}(\mathbf{x}_A | \mathbf{x}_B) = \frac{p((\mathbf{x}_A, \mathbf{x}_B))}{p_{\mathbf{X}_B}(\mathbf{x}_B)}.$$

Example: $f(x_1, x_2) = \frac{1}{2}x_1 + \frac{3}{2}x_2$ if $0 \leq x_1, x_2 \leq 1$ and zero otherwise. The marginal pdf of X_1 and the conditional pdf $X_1 | X_2$ are

$$f_{X_1}(x_1) = \int_0^1 f(x_1, x_2) dx_2 = \frac{1}{2}x_1 + \frac{3}{4}$$

$$f_{X_2 | X_1}(x_2 | x_1) = \frac{\frac{1}{2}x_1 + \frac{3}{2}x_2}{\frac{1}{2}x_1 + \frac{3}{4}}$$

Expectations

The expectation of a continuous random vector $\mathbf{X} = (X_1, \dots, X_d)^\top$ is the d -vector of marginal expectation:

$$\mathbb{E}(\mathbf{X}) = \begin{pmatrix} \mathbb{E}(X_1) \\ \vdots \\ \mathbb{E}(X_d) \end{pmatrix} = \begin{pmatrix} \int_{-\infty}^{+\infty} x f_1(x) dx \\ \vdots \\ \int_{-\infty}^{+\infty} x f_d(x) dx \end{pmatrix}$$

The expectation of a discrete random vector $\mathbf{X} = (X_1, \dots, X_d)^\top$ is the d -vector of marginal expectation:

$$\mathbb{E}(\mathbf{X}) = \begin{pmatrix} \mathbb{E}(X_1) \\ \vdots \\ \mathbb{E}(X_d) \end{pmatrix} = \begin{pmatrix} \sum_{j=1}^{m_1} x_{1,j} p_{j,\bullet,\dots,\bullet} \\ \vdots \\ \sum_{j=1}^{m_d} x_{d,j} p_{\bullet,\dots,\bullet,j} \end{pmatrix}$$

Properties (proofs as exercise):

$$\text{linearity : } \mathbb{E}(\alpha \mathbf{X} + \beta \mathbf{Y}) = \alpha \mathbb{E}(\mathbf{X}) + \beta \mathbb{E}(\mathbf{Y})$$

$$\text{Indep. } \mathbf{X} \text{ and } \mathbf{Y} : \mathbb{E}(\mathbf{X}^\top \mathbf{Y}) = \mathbb{E}(\mathbf{X})^\top \mathbb{E}(\mathbf{Y})$$

Conditional expectations

The expectation of a function $g : \mathbb{R}^d \rightarrow \mathbb{R}$ of the continuous random vector $\mathbf{X} = (X_1, \dots, X_d)^\top$ is the following integral

$$\mathbb{E}(g(\mathbf{X})) = \begin{cases} \int_{-\infty}^{+\infty} \dots \int_{-\infty}^{+\infty} g(u_1, \dots, u_d) f(u_1, \dots, u_d) du_1 \dots du_d \\ \sum_{i_d=1}^{m_d} \dots \sum_{i_1=1}^{m_1} g(x_{1,i_1}, \dots, x_{d,i_d}) p(x_{1,i_1}, \dots, x_{d,i_d}) \end{cases}$$

Warning $\mathbb{E}(g(\mathbf{X})) \neq g(\mathbb{E}(\mathbf{X}))$ unless $g(\cdot)$ is linear.

Let $\mathbf{X} = (X_1, \dots, X_d)^\top$ be a continuous or discrete random vector. We partition the vector into two disjoint sub-vectors $\mathbf{X}_A$ and $\mathbf{X}_B$. The **conditional expectation** of $\mathbf{X}_A$ given $\mathbf{X}_B = \mathbf{x}_B$ is

$$\begin{aligned} \mathbb{E}(\mathbf{X}_A | \mathbf{X}_B = \mathbf{x}_B) &= \int_{\mathbf{u} \in A} \mathbf{u} f_{\mathbf{X}_A | \mathbf{X}_B}(\mathbf{u} | \mathbf{x}_B) d\mathbf{u}, \\ \mathbb{E}(\mathbf{X}_A | \mathbf{X}_B = \mathbf{x}_B) &= \sum_{\mathbf{u} \in A} \mathbf{u} p_{\mathbf{X}_A | \mathbf{X}_B}(\mathbf{u} | \mathbf{x}_B). \end{aligned}$$

Conditional expectations

Let $\mathbf{X}_A$ and $\mathbf{X}_B$ be a partition of the rand. vector $\mathbf{X} = (X_1, \dots, X_d)^\top$.
We have

$$\mathbb{E}(\mathbf{X}_A) = \mathbb{E}(\mathbb{E}(\mathbf{X}_A | \mathbf{X}_B)) . \quad (1)$$

Proof As $f_{\mathbf{X}_A | \mathbf{X}_B}(\mathbf{u} | \mathbf{X}_B = \mathbf{v}) f_{\mathbf{X}_B}(\mathbf{v}) = f(\mathbf{u}, \mathbf{v})$

$$\begin{aligned} \mathbb{E}(\mathbb{E}(\mathbf{X}_A | \mathbf{X}_B)) &= \int_{\mathbf{v} \in B} \left(\int_{\mathbf{u} \in A} \mathbf{u} f_{\mathbf{X}_A | \mathbf{X}_B}(\mathbf{u} | \mathbf{X}_B = \mathbf{v}) d\mathbf{u} \right) f_{\mathbf{X}_B}(\mathbf{v}) d\mathbf{v} \\ &= \int_{\mathbf{u} \in A} \mathbf{u} \int_{\mathbf{v} \in B} f(\mathbf{u}, \mathbf{v}) d\mathbf{v} d\mathbf{u} \end{aligned}$$

As $f_{\mathbf{X}_A}(\mathbf{u}) = \int_{\mathbf{v} \in B} f(\mathbf{u}, \mathbf{v}) d\mathbf{v}$, we conclude. **End**

The conditional expectation $\mathbb{E}(\mathbf{X}_A | \mathbf{X}_B)$ is also the best approximation of $\mathbf{X}_A$, observing $\mathbf{X}_B$ in the quadratic sense.

Expectations

Let $\mathbf{X}_A$ and $\mathbf{X}_B$ be a partition of $\mathbf{X} = (X_1, \dots, X_d)^\top$. The function $g(\cdot)$ solution of

$$\min_g \mathbb{E} \left((\mathbf{X}_A - g(\mathbf{X}_B))^2 \right),$$

is $g(\mathbf{X}_B) = \mathbb{E}(\mathbf{X}_A | \mathbf{X}_B)$.

Proof Using (1), we infer that

$$\begin{aligned} \mathbb{E} \left((\mathbf{X}_A - g(\mathbf{X}_B))^2 \right) &= \mathbb{E} \left(\mathbb{E} \left((\mathbf{X}_A - g(\mathbf{X}_B))^2 | \mathbf{X}_B \right) \right) \\ &= \int_B \mathbb{E} \left((\mathbf{X}_A - g(\mathbf{x}_B))^2 | \mathbf{X}_B = \mathbf{x}_B \right) f_{\mathbf{X}_B}(\mathbf{x}_B) d\mathbf{x}_B \end{aligned}$$

Let us denote $\mathbf{y} = g(\mathbf{x}_B)$ (not random) and

$$L(\mathbf{y}) = \mathbb{E} \left((\mathbf{X}_A - \mathbf{y})^2 | \mathbf{X}_B = \mathbf{x}_B \right).$$

Covariance

We differentiate w.r.t. $\mathbf{y}$ and cancel the gradient $\nabla_{\mathbf{y}} L(\mathbf{y})$:

$$\begin{aligned}\nabla_{\mathbf{y}} L(\mathbf{y}) &= 2\mathbb{E}((\mathbf{X}_A - \mathbf{y}) | \mathbf{X}_B = \mathbf{x}_B) \\ &= 2\mathbb{E}(\mathbf{X}_A | \mathbf{X}_B = \mathbf{x}_B) - 2\mathbf{y} = 0\end{aligned}$$

and then $\mathbf{y} = g(\mathbf{x}_B) = \mathbb{E}(\mathbf{X}_A | \mathbf{X}_B = \mathbf{x}_B)$. **End**

If we denote by $\boldsymbol{\mu} = (\mu_1, \dots, \mu_d)^\top = \mathbb{E}(\mathbf{X})$ the d -vector of expectation, the **covariance matrix** of $\mathbf{X}$ is a $d \times d$ -matrix,

$$\begin{aligned}\Sigma &= \mathbb{E}\left((\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top\right) \\ &= \mathbb{E}\left(\mathbf{X}\mathbf{X}^\top\right) - \boldsymbol{\mu}\boldsymbol{\mu}^\top\end{aligned}$$

Covariance

Let $\mathbf{X}$ be a d -random vector with expectation $\boldsymbol{\mu} = \mathbb{E}(\mathbf{X})$ and covariance matrix $\Sigma_{\mathbf{X}}$. Let M be a $(p \times d)$ -matrix and $\mathbf{m}$ a p -vector. Then $\mathbf{Y} = M\mathbf{X} + \mathbf{m}$ is a p -random vector such that

$$\boldsymbol{\mu}_{\mathbf{Y}} = M\boldsymbol{\mu} + \mathbf{m}$$

$$\Sigma_{\mathbf{Y}} = M\Sigma_{\mathbf{X}}M^{\top}$$

Proof The first statement is a direct consequence of the linearity of the integral. The covariance matrix of $\mathbf{Y}$ is:

$$\begin{aligned}\Sigma_{\mathbf{Y}} &= \mathbb{E}(\mathbf{Y}\mathbf{Y}^{\top}) - \boldsymbol{\mu}_{\mathbf{Y}}\boldsymbol{\mu}_{\mathbf{Y}}^{\top} \\ &= \mathbb{E}\left((M\mathbf{X} + \mathbf{m})(\mathbf{X}^{\top}M^{\top} + \mathbf{m}^{\top})\right) - (M\boldsymbol{\mu} + \mathbf{m})(M\boldsymbol{\mu} + \mathbf{m})^{\top} \\ &= \mathbb{E}\left(M\mathbf{X}\mathbf{X}^{\top}M^{\top}\right) = M\Sigma_{\mathbf{X}}M^{\top}.\end{aligned}$$

Independence

We can show that $\Sigma = (\sigma_{i,j})_{i,j=1,\dots,d}$ is filled with

$$\begin{aligned}\sigma_{i,j} &= \mathbb{C}(X_i, X_j) = \int \int (x_i - \mu_i)(x_j - \mu_j) f_{X_i, X_j}(x_i, x_j) dx_i dx_j \\ \sigma_{i,i} &= \mathbb{V}(X_i) = \int (x_i - \mu_i)^2 f_{X_i}(x_i) dx_i\end{aligned}$$

Let $\mathbf{X} = (X_1, \dots, X_d)^\top$ be a random vector. The r.v. X_k for $k = 1, \dots, d$ are **independent if and only if** the pdf (pmf) is the product of marginal pdfs (pmfs)

$$\begin{aligned}f(x_1, \dots, x_d) &= \prod_{k=1}^d f_{X_k}(x_k). \\ p(x_1, \dots, x_d) &= \prod_{k=1}^d P(X_k = x_k).\end{aligned}$$

Independence

Consequence 1: Let $\mathbf{X}_A$ and $\mathbf{X}_B$ be a partition of $\mathbf{X} = (X_1, \dots, X_d)^\top$, with independent r.v. then

$$\mathbb{E}(\mathbf{X}_A | \mathbf{X}_B = x_B) = \mathbb{E}(\mathbf{X}_A).$$

Reason:

$$f_{\mathbf{X}_A | \mathbf{X}_B}(\mathbf{x}_A | \mathbf{x}_B) = \frac{f(\mathbf{x}_A, \mathbf{x}_B)}{f_{\mathbf{X}_B}(\mathbf{x}_B)} = \frac{f(\mathbf{x}_A)f(\mathbf{x}_B)}{f_{\mathbf{X}_B}(\mathbf{x}_B)} = f(\mathbf{x}_A).$$

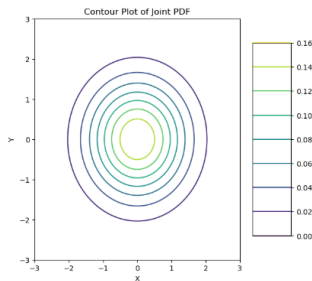
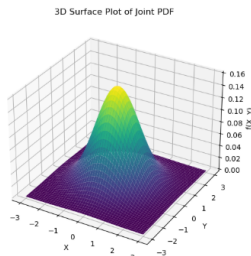
Consequence 2: The covariance matrix is diagonal since $f_{X_i, X_j}(x_i, x_j) = f_{X_i}(x_i)f_{X_j}(x_j)$:

$$\begin{aligned}\sigma_{i,j} &= \int \int (x_i - \mu_i)(x_j - \mu_j) f_{X_i, X_j}(x_i, x_j) dx_i dx_j \\ &= \underbrace{\int (x_i - \mu_i) f_{X_i}(x_i) dx_i}_{=0} \underbrace{\int (x_j - \mu_j) f_{X_j}(x_j) dx_j}_{=0}\end{aligned}$$

Independence

Example : Let $X \sim \mathcal{N}(0, 1)$ and $Y \sim \mathcal{N}(0, 1)$ and assume that X and Y are independent. The joint pdf is

$$f(x, y) = f_X(x)f_Y(y) = \frac{1}{2\pi}e^{-\frac{1}{2}(x^2+y^2)}.$$



Multivariate normal distribution

A popular multivariate distribution is the normal.

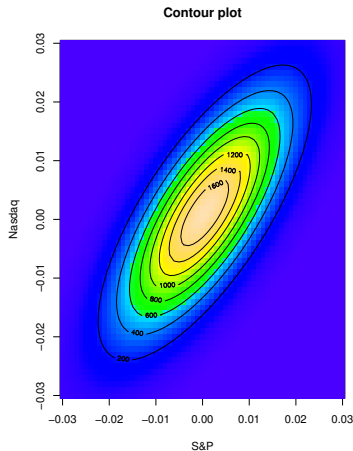
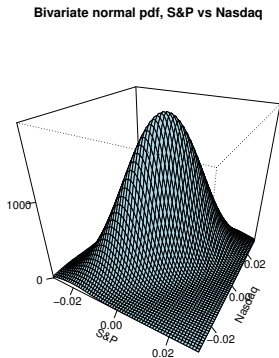
$\mathbf{X} = (X_1, \dots, X_d)$ follows a multivariate **normal distribution** $\mathcal{N}(\boldsymbol{\mu}, \Sigma)$ if its probability density function (pdf) for $\mathbf{x} = (x_1, \dots, x_d) \in \mathbb{R}^d$

$$f(x_1, \dots, x_d) = \frac{1}{\sqrt{\det(2\pi\Sigma)}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})\Sigma(\mathbf{x} - \boldsymbol{\mu})^\top\right)$$

where Σ is the $d \times d$ covariance matrix of $\mathbf{X}$ and $\boldsymbol{\mu} = (\mu_1, \dots, \mu_d)$ is the mean vector.

Bivariate normal distribution

Example: bivariate normal fitted to daily returns of S&P 500 and Nasdaq stocks index.



Multivariate normal distribution

Properties

- ▶ Σ is a diagonal matrix $\Leftrightarrow$ the $X_1, \dots, X_d$ are independent.
- ▶ If $\mathbf{m} \in \mathbb{R}^d$ and $\mathbf{M}$ is a $d \times d$ matrix then

$$\mathbf{Y} = \mathbf{m} + \mathbf{M}\mathbf{X}$$

is a multivariate normal $\mathcal{N}(\boldsymbol{\mu}_Y, \Sigma_Y)$ with mean and covariance:

$$\boldsymbol{\mu}_Y = \mathbf{m} + \mathbf{M}\boldsymbol{\mu}$$

$$\Sigma_Y = \mathbf{M}\Sigma\mathbf{M}^\top$$

Consequences

- ▶ $\mathbf{Y} = \Sigma^{-\frac{1}{2}}\mathbf{X} - \boldsymbol{\mu}$ is $\mathcal{N}(0, I_d)$ (where $\Sigma^{\frac{1}{2}}$ such that $\Sigma^{\frac{1}{2}}\Sigma^{\frac{1}{2}\top} = \Sigma$).
- ▶ $X_j \sim \mathcal{N}(\mu_j, \sigma_{jj})$

Multivariate normal distribution

Conditional MND Let $\mathbf{X} = (\mathbf{X}_A, \mathbf{X}_B)^\top \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$ where $\mathbf{X}_A$ and $\mathbf{X}_B$ are of dimensions p and $d - p$. The expectation and covariance matrix are:

$$\boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu}_A \\ \boldsymbol{\mu}_B \end{pmatrix} \quad \Sigma = \begin{pmatrix} \Sigma_{AA} & \Sigma_{AB} \\ \Sigma_{BA} & \Sigma_{BB} \end{pmatrix}.$$

Define

$$\boldsymbol{\mu}_{B|A}(\mathbf{x}_A) = \boldsymbol{\mu}_B + \Sigma_{BA}\Sigma_{AA}^{-1}(\mathbf{x}_A - \boldsymbol{\mu}_A),$$

$$\Sigma_{B|A}(\mathbf{x}_A) = \Sigma_{BB} - \Sigma_{BA}\Sigma_{AA}^{-1}\Sigma_{AB},$$

then $\mathbf{X}_A \sim \mathcal{N}(\boldsymbol{\mu}_A, \Sigma_{AA})$ and

$$\mathbf{X}_B | \mathbf{X}_A = \mathbf{x}_A \sim \mathcal{N}(\boldsymbol{\mu}_{B|A}(\mathbf{x}_A), \Sigma_{B|A}).$$

Multivariate normal distribution

Example Human height X_1 and weight X_2 form a bivariate random variable $\mathbf{X} = (X_1, X_2)^\top$ with expectation and covariance:

$$\boldsymbol{\mu} = \begin{pmatrix} 70 \text{ kg} \\ 170 \text{ cm} \end{pmatrix} \quad \Sigma = \begin{pmatrix} 80 & 70 \\ 70 & 100 \end{pmatrix}.$$

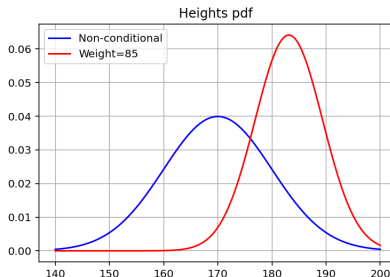
If an individual weights 85kg, what is is expected size? The conditional distribution of size given weight is

$$\begin{aligned} \mu_{B|A}(\mathbf{x}_A) &= 170 + \frac{70}{80} (\mathbf{x}_A - 70), \\ \Sigma_{B|A}(\mathbf{x}_A) &= 100 - \frac{70^2}{80} = (6.22)^2, \end{aligned}$$

The expected height is 183.13 cm with a st. dev. of 6.22 cm.

Multivariate normal distribution

Example Without information about the weight, an individual has an average height of 170 cm with a deviation of 10cm. If we know the weight (85kg), we get a better forecast of the height : average 183 cm with a lower deviation, 6.22cm.



Estimation

Introduction

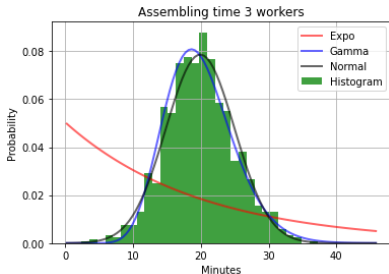
Suppose that a random variable X of interest to an experimenter has a probability distribution with density $f(x|\theta)$ where $\theta \in \Theta$ is a vector of parameters.

To learn about X , the experimenter proceeds by obtaining repeated data (sample values) of the random variable X , written $x_1, x_2, \dots, x_n$. These are the observed values (realizations) of a set of n random variables $X_1, X_2, \dots, X_n$.

The collection of random variables $X_1, X_2, \dots, X_n$ is called a **random sample** (of size n) if (i) they have the same probability distribution $f(x|\theta)$ and (ii) are mutually independent

Introduction

Example 1: 1000 assembling times of a mechanical devices with 3 workers. Each duration is a realization of a random variable. Each measure x_i is the result of a trial X_i , independent from others.



Which candidate distribution for modeling a duration? Compare histogram (emp. pdf) with pdf with the same range: here continuous r.v. : Exponential or Gamma random variables (defined on $\mathbb{R}^+$) or eventually a normal (approximation since the range is $\mathbb{R}$).

Estimator

If we assume that $X_i \sim (\mu, \sigma)$ then $\theta = (\mu, \sigma) \in \Theta = \mathbb{R}_+^2$, **how do we estimate θ ?**

An **estimator** of θ , generically denoted by $\hat{\theta}$ is any function $h(\cdot)$ of the random sample:

$$\hat{\theta} = h(X_1, \dots, X_n) \in \Theta$$

used to estimate θ . As θ is unknown, $\hat{\theta}$ gives an approximation.

An **estimate** of θ , is an observed value of this estimator calculated from the observed sample, $x_1, \dots, x_n$:

$$\hat{\theta}_{obs} = h(x_1, \dots, x_n) \in \Theta$$

Estimator

The estimator is a function of n random variables and **therefore is also a random variable!**

$\hat{\theta}$ is and **unbiased estimator** of θ if

$$\mathbb{E}(\hat{\theta}) = \mathbb{E}(h(X_1, \dots, X_n)) = \theta.$$

The **bias** is the difference between the expectation and the real unknown value:

$$B(\hat{\theta}) = \mathbb{E}(\hat{\theta}) - \theta.$$

The **Mean square error** (MSE) measures the average error:

$$MSE(\hat{\theta}) = \mathbb{E}\left((\hat{\theta} - \theta)^2\right)$$

BIAS-VARIANCE decomposition of the MSE:

$$MSE\left(\hat{\theta}\right) = B(\hat{\theta})^2 + \mathbb{V}\left(\hat{\theta}\right) .$$

Proof:

$$\begin{aligned} MSE\left(\hat{\theta}\right) &= \mathbb{E}\left(\left(\hat{\theta} - \theta\right)^2\right) = \mathbb{E}\left(\hat{\theta}^2\right) - 2\theta\mathbb{E}\left(\hat{\theta}\right) + \theta^2 \\ &= \underbrace{\mathbb{E}\left(\hat{\theta}^2\right) - \mathbb{E}\left(\hat{\theta}\right)^2}_{\mathbb{V}(\hat{\theta})} + \underbrace{\mathbb{E}\left(\hat{\theta}\right)^2 - 2\theta\mathbb{E}\left(\hat{\theta}\right) + \theta^2}_{B(\hat{\theta})^2} \end{aligned}$$

Conclusion: **the best estimator should have the lowest bias and variance!**

Method of Moments

Moments matching

We observe $x_1, \dots, x_n$, realizations of $X_{i=1:n}$. We think that $X_{i=1:n}$ have the same pdf $f(x|\theta)$ as X . In order to estimate $\theta \in \mathbb{R}^d$, we match the d moments

$$\mu_k(\theta) = \mathbb{E} \left((X)^k \right) \quad k = 1, \dots, d$$

with the d empirical moments (r.v. noted M_k , realizations: m_k)

$$M_k = \frac{1}{n} \sum_{i=1}^n X_i^k \quad k = 1, \dots, d$$

The estimator of moments $\hat{\theta}$ is solution of a system with d equations:

$$\begin{aligned} \mu_k(\hat{\theta}) &= M_k & k = 1, \dots, d. \\ \mu_k(\hat{\theta}_{obs}) &= m_k & k = 1, \dots, d. \end{aligned}$$

Moments matching

Example 1: we observe $x_1, \dots, x_n$ realizations of $X \sim \text{expo}(\beta)$, i.e. pdf $f(x|\beta) = \frac{1}{\beta} e^{-\frac{1}{\beta}x}$. According to Appendix 1, $\mathbb{E}(X) = \beta$ and $M_1 = \bar{X} = \frac{1}{n} \sum_{i=1:n} X_i$ then

$$\hat{\beta} = \bar{X} \quad , \quad \hat{\beta}_{obs} = \bar{x}.$$

Example 2: we observe $x_1, \dots, x_n$ realizations of $X \sim \mathcal{N}(\mu, \sigma^2)$ with $\mathbb{E}(X) = \mu$ and $\mathbb{V}(X) = \sigma^2$. If there are only 2 parameters, we match the empirical mean and variance with theoretical mean and variance.

$$\hat{\mu} = \bar{X} \quad , \quad \hat{\mu}_{obs} = \bar{x}.$$
$$\hat{\sigma}^2 = S^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1} \quad , \quad \hat{\sigma}_{obs}^2 = s^2.$$

Example 3: we observe $x_1, \dots, x_n$ realizations of $X \sim \text{Gamma}(\alpha, \beta)$, $\mathbb{E}(X) = \alpha\beta$ and $\mathbb{V}(X) = \alpha\beta^2$.

$$\begin{cases} \hat{\alpha}\hat{\beta} = \bar{X} \\ \hat{\alpha}\hat{\beta}^2 = S^2 \end{cases} \iff \begin{cases} \hat{\alpha} = \frac{\bar{X}^2}{S^2} \\ \hat{\beta} = \frac{S^2}{\bar{X}} \end{cases}$$

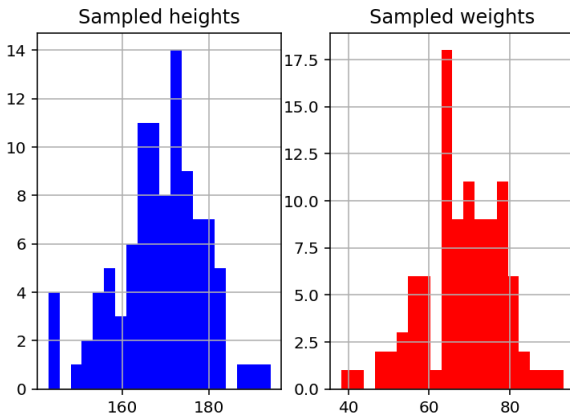
For a Python illustration, see the file “[MomentMatching.py](#)”

Example 4: we perform n experiments, success $X_i = 1$ with proba p , otherwise $X_i = 0$. Realizations of $X \sim \text{Be}(p)$, $\mathbb{E}(X) = p$ then:

$$\hat{p} = \bar{X} \quad , \quad \hat{p}_{obs} = \bar{x}.$$

Moments matching

Example 5: You collect 100 measures of students' heights and weights. You want to fit a bivariate normal distribution.



Moments matching

You match the expectations with averages and the covariance with empirical covariances

$$\begin{pmatrix} \hat{\mu}_H \\ \hat{\mu}_W \end{pmatrix} = \begin{pmatrix} \bar{X}_H \\ \bar{X}_W \end{pmatrix} = \begin{pmatrix} 168.63 \\ 68.27 \end{pmatrix}$$

$$\hat{\Sigma} = \begin{pmatrix} \hat{\sigma}_{HH} & \hat{\sigma}_{HW} \\ \hat{\sigma}_{WH} & \hat{\sigma}_{WW} \end{pmatrix} = \begin{pmatrix} s_{HH} & s_{HW} \\ s_{WH} & s_{WW} \end{pmatrix} = \begin{pmatrix} 99.95 & 79.11 \\ 79.11 & 102.11 \end{pmatrix}$$

You do the same experiment, with this time 1000 observations and you get

$$\begin{pmatrix} \hat{\mu}_H \\ \hat{\mu}_W \end{pmatrix} = \begin{pmatrix} \bar{X}_H \\ \bar{X}_W \end{pmatrix} = \begin{pmatrix} 169.79 \\ 69.83 \end{pmatrix}$$

$$\hat{\Sigma} = \begin{pmatrix} \hat{\sigma}_{HH} & \hat{\sigma}_{HW} \\ \hat{\sigma}_{WH} & \hat{\sigma}_{WW} \end{pmatrix} = \begin{pmatrix} s_{HH} & s_{HW} \\ s_{WH} & s_{WW} \end{pmatrix} = \begin{pmatrix} 96.19 & 69.83 \\ 96.19 & 65.60 \end{pmatrix}$$

Moments matching

Question: are these estimates reliable? We see that they are based on $\bar{X}$ and S^2 . First answer:

$\bar{X}$ is an unbiased estimator of $\mathbb{E}(X)$.

Proof: evident since $\mathbb{E}(\bar{X}) = \frac{1}{n} \sum_{i=1}^n \mathbb{E}(X) = \mathbb{E}(X)$. Therefore in examples 1.1 and 1.2, $\hat{\beta}$ and $\hat{\mu}$ are unbiased estimators.

Second answer: based on the CLT stating that whatever the distribution of $X_{i=1:n} \sim X$,

$$\bar{X} \sim \mathcal{N}\left(\mathbb{E}(X), \frac{\mathbb{V}(X)}{n}\right)$$

then the higher is n , the lower is the variance of $\bar{X}$ around $\mathbb{E}(X)$ (but $\mathbb{E}(X)$ and $\mathbb{V}(X)$ are unknown in practice)

Moments matching

We cannot say a lot about the properties of the moment estimators. In general they are "consistent", i.e. the value of the estimator $\hat{\theta} = h(X_1, \dots, X_n)$ converges to the true value when $n \rightarrow \infty$.

We do not know much about their variance, except that their variance is larger than or equal to the variance of the maximum likelihood estimators (introduced later).

To summarize, the moment estimators are easy to construct but do not always possess the best statistical properties.

A much better approach than moment matching to find an estimator is the likelihood maximization.

Likelihood Maximization

Likelihood Maximization

Let us consider a random sample $X_{1:n} \sim X$. We think that X has a pdf $f(x|\theta)$ where θ is the vector of unknown parameters. Let us recall that

$$P(x \leq X \leq x + dx) \approx f(x|\theta) dx.$$

Since the $X_{i:n}$ are independent, the probability to observe realizations $x_{1:n}$ of $X_{1:n}$ is:

$$\begin{aligned} &P(x_1 \leq X_1 \leq x_1 + dx, \dots, x_n \leq X_n \leq x_n + dx) \\ &= P(x_1 \leq X_1 \leq x_1 + dx) \dots P(x_n \leq X_n \leq x_n + dx) \\ &= \prod_{k=1}^n f(x_k|\theta) dx. \end{aligned}$$

Likelihood Maximization

The probability that the observed sample has been generated by the model is then proportional ($\propto$) to the **likelihood function**:

$$L(x_1, \dots, x_n | \theta) := \prod_{k=1}^n f(x_k | \theta)$$

The maximum likelihood estimator (**MLE**) of θ is the value which maximises the likelihood of the observed sample

$$\hat{\theta} = \arg \max_{\theta} L(x_1, \dots, x_n | \theta)$$

In practice $\hat{\theta}$ is found by deriving w.r.t. θ the log-likelihood function $l(.) = \ln L(.)$ i.e.

$$l(x_1, \dots, x_n | \theta) = \sum_{k=1}^n \ln (f(x_k | \theta)) .$$

Likelihood Maximization

Example 1: If $X_{i:n}$ are $\text{expo}(\beta)$ then $f(x|\beta) = \frac{1}{\beta} e^{-\frac{1}{\beta}x}$ and

$$L(\beta) = \beta^{-n} e^{-\frac{1}{\beta}(x_1 + \dots + x_n)}$$

$$l(\beta) = -n \ln(\beta) - \frac{1}{\beta}(x_1 + \dots + x_n)$$

Then

$$\frac{\partial l}{\partial \beta} = -n \frac{1}{\beta} + \frac{1}{\beta^2}(x_1 + \dots + x_n) = 0$$

And we retrieve here the estimator by moment matching (not always the case!):

$$\hat{\beta}_{obs} = \bar{x} \quad , \quad \hat{\beta} = \bar{X}.$$

When we test several models, we select the one with the highest log-likelihood.

Statistical properties of the MLE :

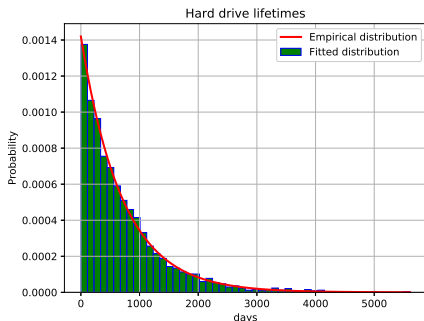
- ▶ A MLE is asymptotically without bias, asymptotically normal.
- ▶ A MLE is of minimum variance :

$$\sqrt{n} \left(\hat{\theta} - \theta \right) \rightarrow \mathcal{N}(0, v(\theta))$$

with asymptotic variance $v(\theta) = \lim_{n \rightarrow \infty} \mathbb{V} \left(\sqrt{n} \hat{\theta} \right)$ to be shown smallest possible of all (asymptotically) unbiased estimators of θ .

Likelihood Maximization

Example 1: see file **bootstrapping.py**. We have a dataset about 2987 lifetimes of hard-drives (in days). We want to model the HDD lifetime by an exponential distribution, $\text{expo}(\beta)$.



We use the scipy command `expon.fit(data=..., scale=...)`. Python finds $\hat{\beta}$ by ML and a localization parameter (loc) such that $X - \text{loc} \sim \text{Expo}(\beta)$. We find $\beta = 704$, loc=0 and the log-likelihood is -22573 .

Likelihood Maximization, discrete r.v.

The maximum likelihood estimator (**MLE**) for a **discrete random variable**

$$\hat{\theta} = \arg \max_{\theta} l(x_1, \dots, x_n | \theta)$$

where $l(\cdot)$ is the sum of log of **pmf** :

$$l(x_1, \dots, x_n | \theta) = \sum_{i=1}^n \ln(p(x_i | \theta)) .$$

Example 2: If $X_{i:n} \in \{0, 1\}$ are $Ber(p)$, $P(X = x) = p^x(1-p)^{1-x}$ and

$$l(p) = \sum_{i=1}^n \ln(p^{x_i}(1-p)^{1-x_i})$$

Since $\frac{\partial l}{\partial p} = \frac{1}{p} \sum_{i=1}^n x_i - (n - \sum_{i=1}^n x_i) \frac{1}{1-p} = 0$ then $\hat{p} = \frac{\sum_{i=1}^n x_i}{n}$.

Likelihood Maximization

For discrete r.v. we replace the pdf by the pmf.

Example 3: If $X_{i:n}$ are $Po(\lambda)$ then $f(x|\lambda) = e^{-\lambda} \frac{\lambda^x}{x!}$ and

$$L(\lambda) = e^{-n\lambda} \frac{\lambda^{(x_1 + \dots + x_n)}}{x_1! \dots x_n!}$$

$$l(\lambda) = -n\lambda - (x_1 + \dots + x_n) \ln \lambda - \ln(x_1! \dots x_n!)$$

Then

$$\frac{\partial l}{\partial \lambda} = -n + \frac{1}{\lambda} (x_1 + \dots + x_n) = 0$$

and

$$\hat{\lambda}_{obs} = \bar{x} \quad , \quad \hat{\lambda} = \bar{X}.$$

Likelihood Maximization

Example 4: If $X_{i:n}$ are distributed according a normal $\mathcal{N}(\mu, \sigma)$ then

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2} (x - \mu)^2\right).$$

The log-likelihood are

$$l(\mu, \sigma) = -\frac{n}{2} \ln(2\pi) - n \ln \sigma - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

Then

$$\frac{\partial l}{\partial \mu} \propto \sum_{i=1}^n (x_i - \mu) = 0$$

$$\frac{\partial l}{\partial \sigma} = -\frac{n}{\sigma} + \frac{1}{\sigma^3} \sum_{i=1}^n (x_i - \mu)^2 = 0$$

Likelihood Maximization

We find that $\hat{\mu} = \bar{X}$ and

$$\hat{\sigma}_{LL}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{\mu})^2 .$$

We see that $\hat{\sigma}_{LL}^2 \neq S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$. In fact, $\hat{\sigma}_{LL}^2$ is biased, we will prove in a following section that

$$\mathbb{E}(\hat{\sigma}_{LL}^2) = \frac{n-1}{n} \sigma^2 ,$$

while $\mathbb{E}(S^2) = \sigma^2$. Reason why we estimate the variance with S^2 .

Simulations & Bootstrapping

Simulations of random numbers

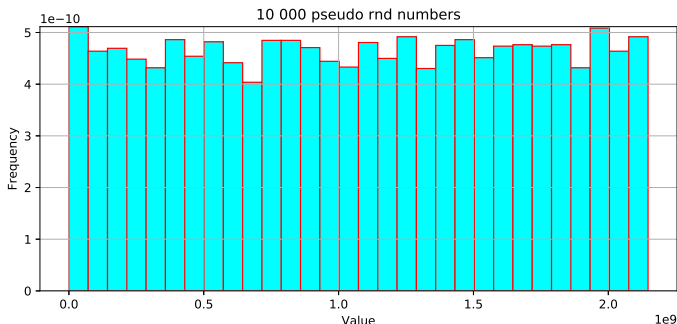
- ▶ In many circumstances, we want to simulate complex system subject to randomness. We therefore need to generate random numbers.
- ▶ A **pseudo-random number generator** (PRNG) is an algorithm for generating a sequence of numbers whose properties approximate the properties of sequences of uniform random numbers. The Linear Congruential Generator (LCG):

$$X_{n+1} = (aX_n + c) \bmod m$$

- ▶ $X = \{X_0, \dots, X_m\}$ is a sequence of pseudo random numbers $\in [0, m]$
 - ▶ m is the modulus (Ansi C/C++ 2^{31})
 - ▶ a is the multiplier (Ansi C/C++ 1103515245)
 - ▶ c is the increment (Ansi C/C++ 12345)
 - ▶ X_0 is the **seed** value

Simulation of random numbers?

- Histogram of 10 000 pseudo-random numbers generated by the LCG algorithm. See file “randomNumbers.py” (e.g. `uniform.rvs(loc=0,scale=1,size=10000)`)



Utility? Simulations of manufacturing chain, financial derivatives pricing, risk management simulations and so on...

Simulation of continuous r.v.

Consider a continuous cdf $F(x)$ which is invertible. Let X be a r.v. defined

$$X = F^{-1}(U)$$

where U is a continuous uniform r.v. on $[0, 1]$. Then X has $F(x)$ for cdf.

Proof:

$$\begin{aligned}P(X \leq x) &= P(F^{-1}(U) \leq x) \\&= P(U \leq F(x)) \\&= F(x)\end{aligned}$$

Then to simulate X , we simulate a sample $\{u_1, \dots, u_n\}$ from $U \sim \text{uni}([0, 1])$ with e.g. the Linear Congruential Generator. A sample of X is then given by computed $\{F^{-1}(u_1), \dots, F^{-1}(u_n)\}$.

Simulation of continuous r.v.

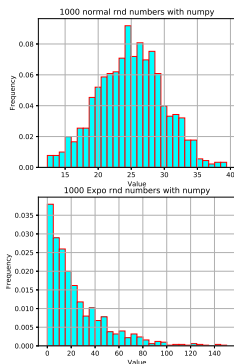
- ▶ In Python, we have the following random numbers generators:

Continuous

- ▶ `uniform.rvs(loc=0,scale=1,size=10000)`
- ▶ `norm.rvs(loc= $\mu$ ,scale= $\sigma$ ,size=10000)`,
- ▶ `t.rvs(df=n,size=10000)`
- ▶ `chi2.rvs(df=n,size=10000)`,
- ▶ `gamma.rvs(a= $\alpha$ ,scale= $\beta$ ,size=10000)`
- ▶ `expon.rvs(scale= $\beta$ , size=1000)`

Discrete

- ▶ `binom.rvs(n=10, p=0.2, size=10000)`
- ▶ `geom.rvs(p=0.2, size=10000)`
- ▶ `poisson.rvs(mu= $\lambda$ , size=10000)`



Bootstrapping

- ▶ The variance of an estimator measures its reliability: the higher is the variance, the lower we should be confident in using this estimate.
- ▶ The variance of an estimator may in some rare (simple) cases be determined analytically.
- ▶ There exists a powerful numerical alternative, called “bootstrapping” and based on resampling.

Bootstrapping

- 1) Resample M new likely data sets from the available one.
- 2) Estimate $\hat{\theta}_k$ for $k = 1, \dots, M$ on each of these data sets.
- 3) Compute the mean and variance of the series of $\left(\hat{\theta}_k\right)_{k=1:M}$.

Bootstrapping

- ▶ Bootstrapping methods are methods based on resampling to substitute complex calculations by Monte Carlo simulations.
- ▶ Let X be a random variable whose the cumulative distribution is noted $F(.)$, which belongs to a parametric family of distribution $\{F_\theta : \theta \in \Theta\}$.
- ▶ We observe a sample of n observations of X , $X = (X_1, \dots, X_n)$. An occurrence of this vector is noted $(x_1, \dots, x_n)$. The empirical cumulative distribution is noted

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n I_{]-\infty, x]}(X_i)$$

- ▶ We denote by $\hat{\theta}(X)$ an estimator of θ , calculated with X .

Bootstrapping

- ▶ A bootstrapped sample is obtained by resampling. Let $X^* = (X_1^* \dots X_n^*)$ such that $P(X_i^* = X_j) = \frac{1}{n}$ for $1 \leq i, j \leq n$. One occurrence $x^* = (x_1^*, \dots, x_n^*)$ is a sequence of n draws from the initial sample with replacement.
 - ▶ $\hat{\theta}(X^*)$ is called **bootstrap replication** of $\hat{\theta}(X)$. The number of possible bootstrap sample is equal to $C_{2n-1}^n = \frac{(2n-1)!}{n!(n-1)!}$.
-
- ▶ Variance is obtained by drawing M bootstrapped sample:
 $\mathbf{X}^{*m} = (X_1^{*m}, \dots, X_n^{*m}) \quad m = 1 \dots M$.

Bootstrapping

θ is estimated by

$$\bar{\theta}^* = \frac{1}{M} \sum_{m=1}^M \hat{\theta}(X^{*m})$$

where $\hat{\theta}(X^{*m})$ is the bootstrap replication of $\hat{\theta}(X)$.

The **empirical variance of the estimator** is ;

$$S^2(\hat{\theta}) = \frac{1}{M-1} \sum_{m=1}^M (\hat{\theta}(X^{*m}) - \bar{\theta}^*)^2$$

Bootstrapping

The sampling distribution of the estimator, $H(x) = P(\hat{\theta}(X) \leq x)$, is the empirical cdf of $(\hat{\theta}(X^{*m}))_{m=1, \dots, M}$

$$H^{boot}(x) = \frac{1}{M} \sum_{m=1}^M 1_{\{\hat{\theta}(X^{*m}) \leq x\}}$$

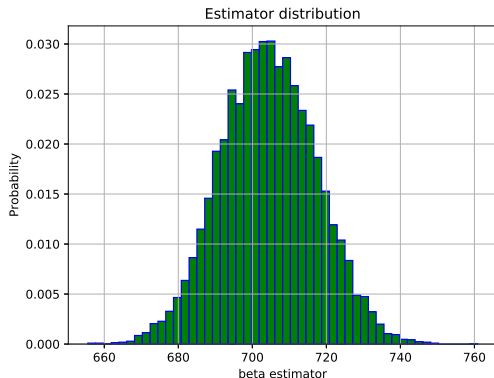
A **confidence interval** at the level α , for θ is obtained from the sampling distribution of the estimator:

$$\left[\underbrace{H_{boot}^{-1}(\alpha/2)}_{\theta_{low}}, \underbrace{H_{boot}^{-1}(1 - \alpha/2)}_{\theta_{up}} \right]. \text{ This is the interval such that:}$$

$$P(\theta_{low} \leq \theta \leq \theta_{up}) = \alpha$$

Bootstrapping

Example : see file **bootstrapping.py**. We have a dataset about 2987 lifetimes of hard-drives (in days). We fit an $\text{expo}(\beta)$.



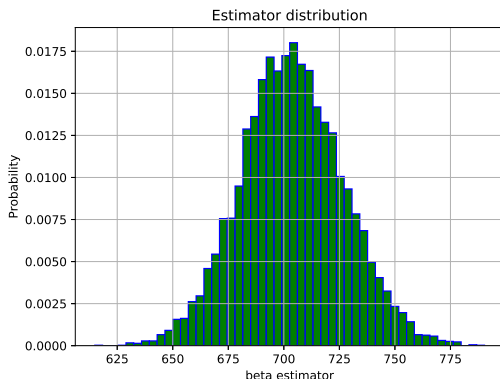
```
Beta estimate :704.38  
Beta standard deviation :12.92  
Beta 2,5% quantile :679.0  
Beta 97,5% quantile :730.0
```

The 95% confidence interval is such that :

$$P(679 \leq \beta \leq 730) = 95\%$$

Bootstrapping

Example (con't): see file **bootstrapping.py**. We use only 1000 records instead of 2987.



```
Beta estimate :703.77
Beta standard deviation :23.05
Beta 2,5% quantile :659.0
Beta 97,5% quantile :750.0
```

Less data... then more uncertainty about the real value of β . The 95% confidence interval is bigger than with the full dataset:

$$P(659 \leq \beta \leq 750) = 95\%$$

Empirical mean and standard deviations: properties

Properties of $\bar{X}$ and S^2

Are estimators reliable ? We have seen that most of them use the empirical mean $\bar{X}$ and variance S^2 . What are the properties of these statistics?

Of course, we can use the bootstrapping approach for valuing their variances (=accuracy) but in many circumstances, the statistical distribution of $\bar{X}$ and S^2 is known.

Let us consider n random variables $X_{i=1:n} \sim X$ and denote $\mu = \mathbb{E}(X)$ and $\sigma^2 = \mathbb{V}(V)$. The CLT states that whatever the distribution of $X_{i=1:n}$, the empirical mean tends to a normal

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \quad \frac{\bar{X} - \mu}{\sqrt{\sigma^2/n}} \sim N(0, 1)$$

Properties of $\bar{X}$ and S^2

A Confidence interval for μ at level $1 - \alpha$ (e.g. $\alpha=5\%$) is an interval $[\mu_L, \mu_U]$ such that μ is in this interval with a probability $1 - \alpha$.

In this case, $\bar{X}$ is an estimator of μ and σ^2 is known. Since $\frac{\bar{X} - \mu}{\sqrt{\sigma^2/n}} \sim N(0, 1)$ we infer that

$$P\left(z_{\alpha/2} \leq \frac{\bar{X} - \mu}{\sqrt{\sigma^2/n}} \leq z_{1-\alpha/2}\right) = 1 - \alpha$$

But $-z_{1-\alpha/2} = z_{\alpha/2}$. The $1 - \alpha$ confidence interval for μ is then

$$\left[\bar{X} - z_{1-\alpha/2} \sqrt{\sigma^2/n}; \bar{X} + z_{1-\alpha/2} \sqrt{\sigma^2/n} \right]$$

Properties of $\bar{X}$ and S^2

Example 1: a) A sample of $n = 40$ Gamma r.v. X with a mean $\mu = 100$ and variance $\sigma^2 = 25$. What is the probability that the sample mean $\bar{X}$ is between 98.4505 and 101.5495?

$$\begin{aligned} & P(98.4505 \leq \bar{X} \leq 101.5495) \\ & \approx P\left(\frac{98.4505 - 100}{\sqrt{25/40}} \leq Z \leq \frac{101.5495 - 100}{\sqrt{25/40}}\right) \\ & = P(-1.96 \leq Z \leq 1.96) \\ & = P(Z \leq 1.96) - P(Z \leq -1.96) = 0.95 \end{aligned}$$

b) If we do not know μ , can we find a 95% confidence interval ($\alpha = 5\%$) for μ ? If $\bar{X} = 99.89$ and $Z_{97.5\%} = 1.96$ then

$$\mu \in \left[99.89 - 1.96\sqrt{\frac{25}{40}}; 99.89 + 1.96\sqrt{\frac{25}{40}} \right] = [98.34; 101.44]$$

Numerical solution: see Python file [XbarAndS2examples.py](#)

Properties of $\bar{X}$ and S^2

1. $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ is an unbiased estimator of $\mathbb{V}(X)$.
2. If $X_i \sim \mathcal{N}(\mu, \sigma^2)$, S^2 and $\bar{X}$ are independent and then

$$(n-1) \frac{S^2}{\sigma^2} = \sum_{i=1}^n \left(\frac{X_i - \bar{X}}{\sigma} \right)^2 \sim \chi_{n-1}^2 \quad (2)$$

Proof. (for information). We denote by $SST = \sum_{i=1}^n (X_i - \bar{X})^2$, the sum of squared total deviations and $\mathbf{X} = (X_1, \dots, X_n)^\top$. Let $\mathbf{I}_n$ be $n \times n$ identity matrix and $\mathbf{J}_n$ is the $n \times n$ matrix of ones. The SST is rewritten under matrix form:

$$SST = \mathbf{X}^\top \left(\mathbf{I}_n - \frac{1}{n} \mathbf{J}_n \right) \mathbf{X}$$

The matrix $\mathbf{M} = \left(\mathbf{I}_n - \frac{1}{n} \mathbf{J}_n \right)$ is idempotent, $\mathbf{M}^2 = \mathbf{M}$, and symmetric, $\mathbf{M} = \mathbf{M}^\top$.

Properties of $\bar{X}$ and S^2

Proof (cont'd). The trace of $\mathbf{M}$ is equal to $tr(\mathbf{M}) = n - 1$. We next use the properties that eigenvalues of idempotent matrix must be equal to zero or one. To prove this, let ϕ the normalized eigenvector of $\mathbf{M}$ with eigenvalue λ : $\mathbf{M}\phi = \lambda\phi$ then

$$\underbrace{\mathbf{M}\mathbf{M}}_{=\mathbf{M}}\phi = \lambda \underbrace{\mathbf{M}\phi}_{\lambda\phi} \Rightarrow \lambda\phi = \lambda^2\phi$$

and $\lambda = 0$ or 1 . The trace of $\mathbf{M}$ is also the sum of eigenvalues. Then $\mathbf{M}$ has $n - 1$ eigenvalues equal to 1 and one equal to zero. The spectral decomposition of $\mathbf{M}$ is $\mathbf{M} = \mathbf{A}\mathbf{D}\mathbf{A}^\top$ where $\mathbf{A}\mathbf{A}^\top = \mathbf{I}_n$ because $\mathbf{M}$ is symmetric and

$$\mathbf{D} = \begin{pmatrix} \mathbf{I}_{n-1} & \mathbf{0}_{(n-1) \times 1} \\ \mathbf{0}_{1 \times (n-1)} & 0 \end{pmatrix}$$

We infer that $(\mathbf{D}\mathbf{A}^\top\mathbf{X})_n = 0$. Since $\mathbf{D} = \mathbf{D}^\top$ and $\mathbf{D} = \mathbf{D}\mathbf{D}$:

$$SST = \mathbf{X}^\top \mathbf{M} \mathbf{X} = (\mathbf{X}^\top \mathbf{A} \mathbf{D}) (\mathbf{D}^\top \mathbf{A}^\top \mathbf{X})$$

Properties of $\bar{X}$ and S^2

Proof (cont'd). We note $\mathbf{e} = (1, \dots, 1)^\top$. Since $\mathbf{X} \sim \mathcal{N}(\mu\mathbf{e}, \sigma^2\mathbf{I}_n)$, using standard normal theory:

$$\mathbf{D}^\top \mathbf{A}^\top \mathbf{X} \sim \mathcal{N}(\mu \mathbf{D}^\top \mathbf{A}^\top \mathbf{e}, \sigma^2 \mathbf{D}^\top \mathbf{A}^\top \mathbf{A} \mathbf{D}) \sim \mathcal{N}(\mu \mathbf{D}^\top \mathbf{A}^\top \mathbf{e}, \sigma^2 \mathbf{D})$$

showing that components of $\mathbf{D}^\top \mathbf{A}^\top \mathbf{X}$ with $(\mathbf{D}^\top \mathbf{A}^\top \mathbf{X})_i \sim \mathcal{N}(0, \sigma^2)$ for $i = 1, \dots, n-1$ and $(\mathbf{D}^\top \mathbf{A}^\top \mathbf{X})_n = 0$. Then $(\mathbf{X}^\top \mathbf{A} \mathbf{D} \mathbf{A}^\top \mathbf{X}) / \sigma^2$ is χ_{n-1}^2 with expectation equal to $n-1$:

$$\mathbb{E}\left(\left(\mathbf{X}^\top \mathbf{A} \mathbf{D} \mathbf{A}^\top \mathbf{X}\right) / \sigma^2\right) = n-1.$$

Therefore the estimator is unbiased:

$$\mathbb{E}(S^2) = \mathbb{E}\left(\frac{(\mathbf{X}^\top \mathbf{A} \mathbf{D} \mathbf{A}^\top \mathbf{X})}{n-1}\right) = \sigma^2.$$

end

Properties of $\bar{X}$ and S^2

A Confidence interval for σ^2 at level $1 - \alpha$ (e.g. $\alpha=5\%$) is an interval $[\sigma_L^2, \sigma_U^2]$ such that σ^2 is in this interval with a probability $1 - \alpha$.

If $X \sim N(\mu, \sigma^2)$, S^2 is an estimator of σ . Since $(n-1) \frac{S^2}{\sigma^2} \sim \chi_{n-1}^2$ we infer that

$$P\left(\chi_{n-1, \alpha/2}^2 \leq (n-1) \frac{S^2}{\sigma^2} \leq \chi_{n-1, 1-\alpha/2}^2\right) = 1 - \alpha$$

The $1 - \alpha$ confidence interval is then

$$\left[\frac{(n-1)}{\chi_{n-1, 1-\alpha/2}^2} S^2; \frac{(n-1)}{\chi_{n-1, \alpha/2}^2} S^2 \right]$$

Properties of $\bar{X}$ and S^2

Example 2 a) An i.i.d. sample of size $n = 100$ is drawn from a normal population with $\sigma^2 = 25$. If S^2 is used to estimate σ^2 , what is the probability that the absolute error is greater than 5?

$$\begin{aligned}P(|S^2 - \sigma^2| > 5) &= P((S^2 - \sigma^2) > 5) + P((S^2 - \sigma^2) < -5) \\&= P\left(99 \frac{S^2}{\sigma^2} > 99 \left(\frac{5}{\sigma^2} + 1\right)\right) + P\left(99 \frac{S^2}{\sigma^2} < 99 \left(1 - \frac{5}{\sigma^2}\right)\right) \\&= P(\chi_{99}^2 > 118.8) + P(\chi_{99}^2 < 79.2) = 0.1568\end{aligned}$$

b) We do not know σ . What is the 95% conf. interval ($\alpha = 5\%$) for σ ? If $S^2 = 5.22^2$, $\chi_{99, \alpha/2}^2 = 73.36$, $\chi_{99, 1-\alpha/2}^2 = 128.42$:

$$\sigma \in \left[\sqrt{\frac{99}{\chi_{99, 1-\alpha/2}^2} S^2}; \sqrt{\frac{99}{\chi_{99, \alpha/2}^2} S^2} \right] = [4.58; 6.07]$$

Numerical solution: see Python file [XbarAndS2examples.py](#)

Properties of $\bar{X}$ and S^2

A student's t r.v. is $T_n = \frac{Y}{\sqrt{Z/n}}$ where $Z \sim \chi_n^2$ and $Y \sim N(0, 1)$. Therefore:

If $X_i \sim \mathcal{N}(\mu, \sigma^2)$ then the following ratio is a Student's T r.v.

$$\frac{\bar{X} - \mu}{\sqrt{\frac{S^2}{n}}} \sim t_{n-1} \quad (3)$$

Proof: from previous results, $\bar{X} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$ then

$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1)$. Furthermore, $(n-1) \frac{S^2}{\sigma^2} \sim \chi_{n-1}^2$. We conclude that

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} / \left(\sqrt{\frac{(n-1) S^2}{(n-1) \sigma^2}} \right) \sim t_{n-1}.$$

Properties of $\bar{X}$ and S^2

A Confidence interval for μ at level $1 - \alpha$ (e.g. $\alpha=5\%$) is an interval $[\mu_L, \mu_U]$ such that μ is in this interval with a probability $1 - \alpha$.

In this case, $\bar{X}$ is an estimator of μ and σ is unknown:

$$P\left(t_{n-1, \alpha/2} \leq \frac{\bar{X} - \mu}{\sqrt{S^2/n}} \leq t_{n-1, 1-\alpha/2}\right) = 1 - \alpha$$

the Student's T is symmetric: $-t_{n-1, \alpha/2} = t_{n-1, 1-\alpha/2}$. The $1 - \alpha$ confidence interval is then

$$\mu \in \left[\bar{X} - t_{n-1, 1-\alpha/2} \sqrt{S^2/n}; \bar{X} + t_{n-1, 1-\alpha/2} \sqrt{S^2/n} \right]$$

Properties of $\bar{X}$ and S^2

Example 3 An i.i.d. sample of size $n = 100$ is drawn from a normal population. We do not know $\sigma^2 (= 10^2)$ and $\mu (= 100)$.

b) What is the 95% confidence interval ($\alpha = 5\%$) for μ ? If $\bar{X} = 100.91$, $S^2 = (10.97)^2$, $t_{99, 1-\alpha/2} = 1.98$:

$$\mu \in \left[100.91 - 1.98 \frac{10.97}{\sqrt{100}} ; 100.91 + 1.98 \frac{10.97}{\sqrt{100}} \right] = [98.78 ; 102.15]$$

Numerical solution: see Python file [XbarAndS2examples.py](#)

Properties of $\bar{X}$ and S^2

	Distribution of X	μ	σ^2	Statistics
$\bar{X}$	unknown and large n	ok	ok	$\frac{\bar{X}-\mu}{\sqrt{\sigma^2/n}} \sim \mathcal{N}(0, 1)$
S^2	$\mathcal{N}(\mu, \sigma^2)$	-	ok	$(n-1) \frac{S^2}{\sigma^2} \sim \chi^2_{n-1}$
$\bar{X}$	$\mathcal{N}(\mu, \sigma^2)$	ok	-	$\frac{\bar{X}-\mu}{\sqrt{\frac{S^2}{n}}} \sim t_{n-1}$

This table is the key to understand Hypothesis testing.

Properties of $\bar{X}$ and S^2 , 2 populations

Properties of $\bar{X}$ and S^2 , 2 populations

We consider two i.i.d. normal samples:

$$\mathbf{X}_1 = \{X_{1,1}, \dots, X_{1,n_1}\} \sim \mathcal{N}(\mu_1, \sigma_1^2) \Rightarrow \bar{X}_1 \sim \mathcal{N}\left(\mu_1, \frac{\sigma_1^2}{n_1}\right)$$

$$\mathbf{X}_2 = \{X_{2,1}, \dots, X_{2,n_2}\} \sim \mathcal{N}(\mu_2, \sigma_2^2) \Rightarrow \bar{X}_2 \sim \mathcal{N}\left(\mu_2, \frac{\sigma_2^2}{n_2}\right)$$

We note $\bar{X}_1 = \sum_{i=1:n} X_{1,i}/n_1$ and $\bar{X}_2 = \sum_{i=1:n} X_{2,i}/n_2$. If we remember the properties of normal r.v.,

$$\bar{X}_1 - \bar{X}_2 \sim \mathcal{N}\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)$$

and the normalized difference is

$$\frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim \mathcal{N}(0, 1) \quad (4)$$

Properties of $\bar{X}$ and S^2 , 2 populations

Unbiased estimator of σ_1^2 and σ_2^2 are

$$S_1^2 = \frac{1}{n_1 - 1} \sum_{i=1}^n (X_{1,i} - \bar{X}_1)^2 \quad S_2^2 = \frac{1}{n_2 - 1} \sum_{i=1}^n (X_{2,i} - \bar{X}_2)^2$$

The following ratio is a Fisher-Snedecor random variable:

$$\frac{S_1^2 \sigma_2^2}{S_2^2 \sigma_1^2} \sim F_{n_1-1, n_2-1} \quad (5)$$

Proof: $(n_1 - 1) \frac{S_1^2}{\sigma_1^2} \sim \chi_{n_1-1}^2$ and $(n_2 - 1) \frac{S_2^2}{\sigma_2^2} \sim \chi_{n_2-1}^2$ then

$$\frac{\frac{(n_1-1)}{n_1-1} \frac{S_1^2}{\sigma_1^2}}{\frac{(n_2-1)}{n_2-1} \frac{S_2^2}{\sigma_2^2}} \sim F_{n_1-1, n_2-1}$$

Properties of $\bar{X}$ and S^2 , 2 populations

If the two populations have the same variance $\sigma_1^2 = \sigma_2^2 = \sigma^2$ (but $\mu_1 \neq \mu_2$), an unbiased “pooled” estimator of this variance is

$$S_{pool}^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2},$$

i.e. $\mathbb{E}(S_{pool}^2) = \sigma^2$ and

$$(n_1 + n_2 - 2) \frac{S_{pool}^2}{\sigma^2} \sim \chi_{n_1 + n_2 - 2}^2. \quad (6)$$

Proof: exactly the same as the proof for 1 population.

Properties of $\bar{X}$ and S^2 , 2 populations

A student's t r.v. is $T_n = \frac{Z}{\sqrt{Y/n}}$ where $Z \sim \chi_n^2$ and $Y \sim N(0, 1)$. Therefore:

If the two populations have the same variance $\sigma_1^2 = \sigma_2^2 = \sigma^2$ (but $\mu_1 \neq \mu_2$)

$$\frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_{pool} \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{n_1+n_2-2}. \quad (7)$$

Proof: Direct consequence from previous results,
 $\frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim \mathcal{N}(0, 1)$ and $(n_1 + n_2 - 2) \frac{S_{pool}^2}{\sigma^2} \sim \chi_{n_1+n_2-2}^2$.

Properties of $\bar{X}$ and S^2 , 2 normal populations

		μ_1, μ_2	σ_1^2, σ_2^2	Statistics
$\bar{X}_1 - \bar{X}_2$	$\sigma_1^2 \neq \sigma_2^2$	ok	ok	$\frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim \mathcal{N}(0, 1)$
S_{pool}^2	$\sigma_1^2 = \sigma_2^2$	-	ok	$(n_1 + n_2 - 2) \frac{S_{pool}^2}{\sigma^2} \sim \chi_{n_1 + n_2 - 2}^2$
S_1^2, S_2^2	$\sigma_1^2 \neq \sigma_2^2$	-	ok	$\frac{S_1^2 \sigma_2^2}{S_2^2 \sigma_1^2} \sim F_{n_1 - 1, n_2 - 1}$
$\bar{X}_1 - \bar{X}_2$	$\sigma_1^2 = \sigma_2^2$	ok	-	$\frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_{pool} \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{n_1 + n_2 - 2}$

This table is the key to understand Hypothesis testing for 2 populations

Hypothesis testing, 1 population

Definition: a **hypothesis** is a claim (assertion) about a parameter θ of a random sample.

The **null hypothesis** is denoted by $H_0 = \theta \in \Theta_0$ and states the assertion to be tested. The alternative hypothesis is $H_1 = \theta \in \Theta_1$.

Example: 25-30 years users of Facebook spend per day, on average μ_0 hours on the website.

We note by μ the average time per day spent on Facebook for these users, H_0 is

$$H_0 : \mu = \mu_0 \quad H_1 : \mu \neq \mu_0$$

But μ is not observable... we must estimate it .

Introduction

Based on some observations which are sampled from the probability distribution, we will make our decision of accepting or rejecting H_0 . This procedure is called a **statistical test**.

The decision whether to reject the null H_0 is based on the sample. There are 4 possible cases summarized in the following table :

Decision	Reality	
	H_0 is true	H_1 is true
Accept H_0	Correct	Type 2 error
Reject H_0	Type 1 error	Correct

The probability of making a type I error is called the **significance level** of the test and is usually denoted as α : $P(\text{Type 1 error}) = \alpha$

Introduction

H_0 : you are not sick



Type 1 error

Type 2 error

- ▶ Type I : you're not ill but the COVID test is positive (false positive test)
- ▶ Type II : you are ill but the test is negative (false negative test)

Introduction

The decision to reject the null hypothesis is based on a test statistic, noted $T(\cdot)$. Given a sample $\mathbf{X} = \{X_1, \dots, X_n\}$ of observations, $T(\mathbf{X}) : \mathbb{R}^n \rightarrow \mathbb{R}$. $T(\mathbf{X})$ is such that its distribution is known.

- 1) For a chosen α , we determine a rejection region $R_\alpha \subset \mathbb{R}$, such that $P(\text{Type 1 error}) = \alpha$.
- 2) We calculate $t = T(\mathbf{x})$ the observed value of $T(\mathbf{X})$ under H_0 .
- 3) Decision:
 - ▶ If $t \in R_\alpha$ then reject H_0 .
 - ▶ if $t \notin R_\alpha$ then do not reject H_0 .

Single mean test, σ^2 unknown

We consider a i.i.d. sample $X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma^2)$ with unknown variance. We test

$$H_0 : \mu = \mu_0$$

against 3 alternatives:

$$a) \quad H_1 : \mu > \mu_0$$

$$b) \quad H_1 : \mu < \mu_0$$

$$c) \quad H_1 : \mu \neq \mu_0$$

If H_0 is true, from equation (3), a good choice for the test statistic is the Student's t :

$$T(\mathbf{X}) = \frac{\bar{X} - \mu_0}{\sqrt{\frac{S^2}{n}}} \sim t_{n-1}$$

Single mean test, σ^2 unknown

If $t = T(\mathbf{x})$ is the observed value of the Student's statistic is “too far” from the mean of the student's distribution, it is likely that H_0 is false.

We first find a critical value c under the assumption that H_0 is true:

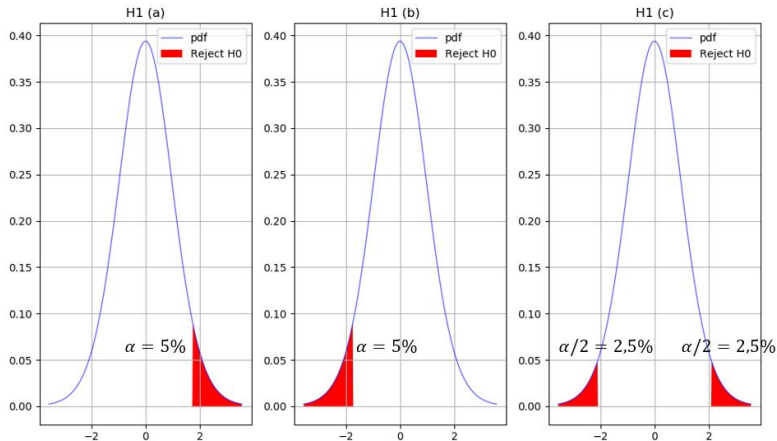
$$a) \quad P(T(\mathbf{X}) > c \mid H_0 \text{ is true}) = \alpha$$

$$b) \quad P(T(\mathbf{X}) < c \mid H_0 \text{ is true}) = \alpha$$

$$c) \quad P(T(\mathbf{X}) < -c \text{ or } T(\mathbf{X}) > c \mid H_0 \text{ is true}) = \alpha$$

The critical values are the α , $1 - \alpha$ or $(\alpha/2, 1 - \alpha/2)$ quantiles of the Student's t (c and $-c$ because t is symmetric)

Single mean test, σ^2 unknown



E.g. : $H_1(a)$, $\mu > \mu_0$: if $T(\mathbf{x})$ is in the red area, a smaller and then more likely statistics maybe obtained with a higher $\mu_0 \Rightarrow$ we reject H_0 .

Single mean test, σ^2 unknown

We reject H_0 at the level α if $T(\mathbf{x}) = \sqrt{n} \frac{\bar{x} - \mu_0}{s}$

a) $H_1 : \mu > \mu_0 \quad T(\mathbf{x}) > t_{n-1} \text{ } 1-\alpha$

b) $H_1 : \mu < \mu_0 \quad T(\mathbf{x}) < t_{n-1} \alpha$

c) $H_1 : \mu \neq \mu_0 \quad T(\mathbf{x}) < t_{n-1} \alpha/2 \text{ or } T(\mathbf{x}) > t_{n-1} 1-\alpha/2$

Single mean test, σ^2 unknown

Example 1. `SampleMeanTest.py` A bottle filling machine is set to fill bottles with soft drink to a volume of 500 ml. The actual volume is known to follow a normal distribution. The manufacturer believes the machine does not work correctly. A sample of 20 bottles is taken and the volume of liquid inside is measured.

test	Volume
1	484.11
	...
	502.85
	...
19	449.08
20	489.27

$H_0 : \mu = 500$ v.s. $H_1 : \mu \neq 500$ Easy to program, package `scipy.stats`:

```
In [44]: Tx= (Stat.mean-500)/np.sqrt(Stat.variance/n)
...: # we compare it to percentiles of a t distribution
...: alpha = 0.05
...: t_l = sc.t.ppf(q=alpha/2,df=n-1)
...: t_u = sc.t.ppf(q=1-alpha/2,df=n-1)
...: #we see that Tx is in the 2.5% and 97.5% interval of the
Student's t
...: print([Tx,t_l,t_u])
[-1.5204626102079255, -2.0930240544082634, 2.093024054408263]
```

alternative: `ttest_1samp()`, We do not reject H_0 for $\alpha = 5\%$.

Single variance test

We consider a i.i.d. sample $X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma^2)$ with unknown variance. We test

$$H_0 : \sigma^2 = \sigma_0^2$$

against 3 alternatives:

$$a) \quad H_1 : \sigma^2 \neq \sigma_0^2$$

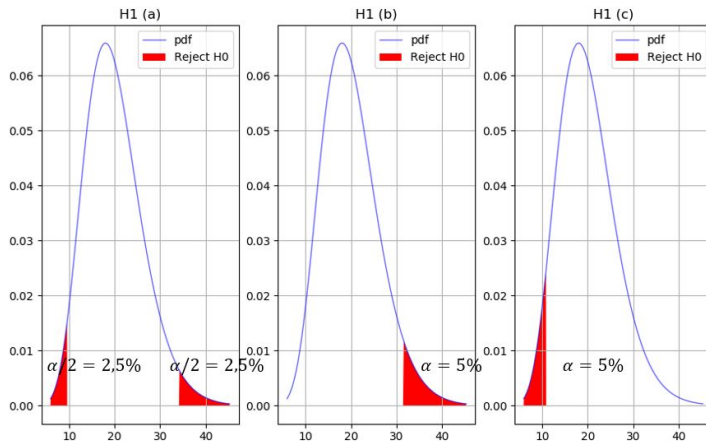
$$b) \quad H_1 : \sigma^2 > \sigma_0^2$$

$$c) \quad H_1 : \sigma^2 < \sigma_0^2$$

From Equation (2), a good choice for the test statistic is the χ^2 test:

$$T(\mathbf{X}) = (n-1) \frac{S^2}{\sigma_0^2} \sim \chi_{n-1}^2$$

Single variance test



E.g. : $H_1(c)$, $\sigma^2 < \sigma_0^2$: if $T(\mathbf{x})$ is in the red area, a higher and then more likely statistics maybe obtained with a smaller $\sigma_0 \Rightarrow$ we reject H_0 .

Single variance test

We reject H_0 at the level α if $T(\mathbf{x}) = (n-1) \frac{s^2}{\sigma_0^2}$

a) $H_1 : \sigma^2 \neq \sigma_0^2$

$$T(\mathbf{x}) < \chi_{n-1, \alpha/2}^2 \text{ or } T(\mathbf{x}) > \chi_{n-1, 1-\alpha/2}^2$$

b) $H_1 : \sigma^2 > \sigma_0^2$

$$T(\mathbf{x}) > \chi_{n-1, 1-\alpha}^2$$

c) $H_1 : \sigma^2 < \sigma_0^2$

$$T(\mathbf{x}) < \chi_{n-1, \alpha}^2$$

Single variance test

Example 1 cont'd. `SampleVarianceTest.py` The quality control dept. requires that the st. dev. of soda volumes is not higher than 20ml to avoid complaints from customers.

One sided sample variance test: $H_0 : \sigma = 20$,
 $H_1 : \sigma > 20$. No function available but easy to program:

test	Volume
1	484.11
	...
	502.85
	...
19	449.08
20	489.27

```
In [2]:
...:
...:
...: sg0=20
...: Tx= (n-1)*Stat.variance/sg0**2
...: # we compare it to percentiles of a chi2 distribution
...: alpha = 0.05
...: t_u = sc.chi2.ppf(q=1-alpha,df=n-1)
...: #we see that Tx is in the 95% interval
...: #of a chi-square
...: print([Tx,t_u])
[29.199522237499995, 30.14352720564616]
```

We do not reject $H_0 : \sigma^2 = 20^2$ for $\alpha = 5\%$

P-value : It is the smallest level of significance for which the data indicate rejection of the null hypothesis.

Let $T(\mathbf{X})$ be a test statistic such that small values of T give evidence that H_0 is wrong. For a given sample $\mathbf{x}$, the p-value is:

$$p(\mathbf{x}) = P(T(\mathbf{X}) < T(\mathbf{x}) \mid H_0 \text{ is true})$$

Let $T(\mathbf{X})$ be a test statistic such that high values of T give evidence that H_0 is wrong. The p-value is:

$$p(\mathbf{x}) = P(T(\mathbf{X}) > T(\mathbf{x}) \mid H_0 \text{ is true})$$

Let $T(\mathbf{X})$ be a test statistic **symmetric around zero** such that high and low values of T give evidence that H_0 is wrong. The p-value is:

$$p(\mathbf{x}) = 2P(T(\mathbf{X}) > |T(\mathbf{x})| \mid H_0 \text{ is true})$$

P-value

A small p-value indicates that H_0 is very unlikely. A high p-value informs us that H_0 is likely.

Example: we consider a i.i.d. sample $X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma^2)$ with unknown variance. We test

$$H_0 : \mu = \mu_0$$

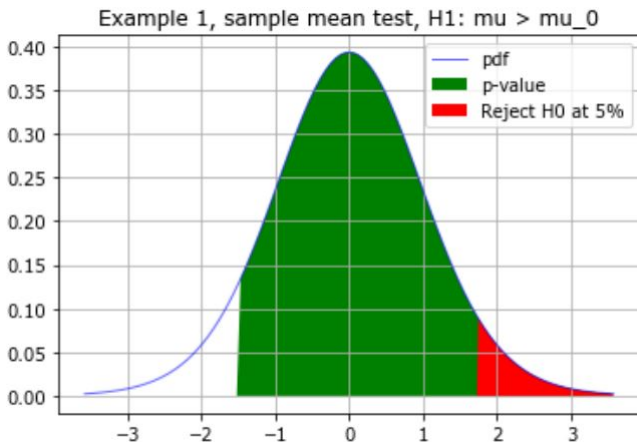
against $H_1 : \mu > \mu_0$. The p-value is such that $p(\mathbf{x}) = P(t_{n-1} > T(\mathbf{x}))$ where $T(\mathbf{x}) = \sqrt{n} \frac{\bar{x} - \mu_0}{s}$. For a confidence level, α , we reject H_0 if

$$p < \alpha \Leftrightarrow T(\mathbf{x}) > t_{n-1, \alpha}.$$

Rejecting a null hypothesis at level α using the critical region method is equivalent to rejecting H_0 when $p(\mathbf{x}) < \alpha$.

P-value

Example 1. $H_0 : \mu = 500$ v.s. $H_1 : \mu > 500$. $T(x) = -1.5204$ and $p = 0.9275$.



Example 1 cont'd. `SampleMeanTest.py` and `SampleVarianceTest.py`.

p-value for the 2 sided T test
($H_1 : \mu \neq 500$) on the average
volume $p = 2 P(t_{n-1} > |T(\mathbf{x})|)$

```
In [264]: pval = 2 * (1-sc.t.cdf(np.abs(Tx),df=n-1))
...: print(pval)
0.14486225283259024
```

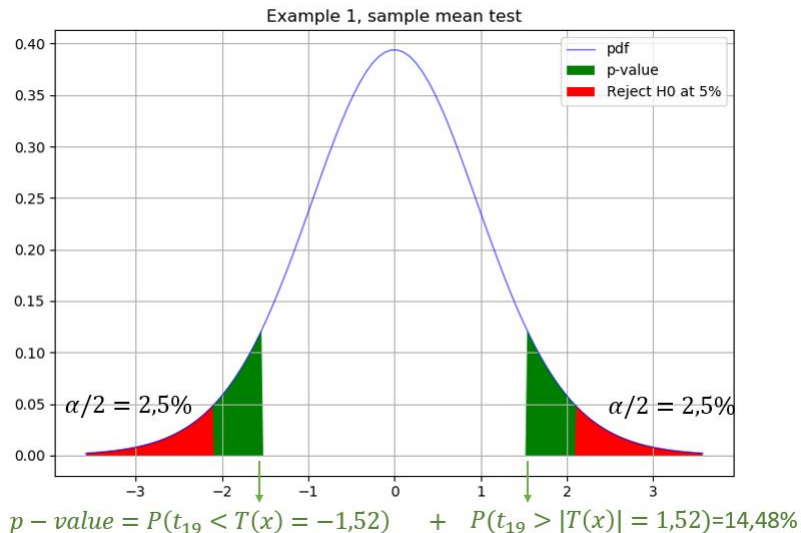
p-value > 5%. We do not reject
 $H_0 : Vol. = 500$ ($\alpha = 5\%$)

p-value for the one sided sample
variance test. p-values:
 $p = P(\chi_{n-1}^2 > T(\mathbf{x}))$

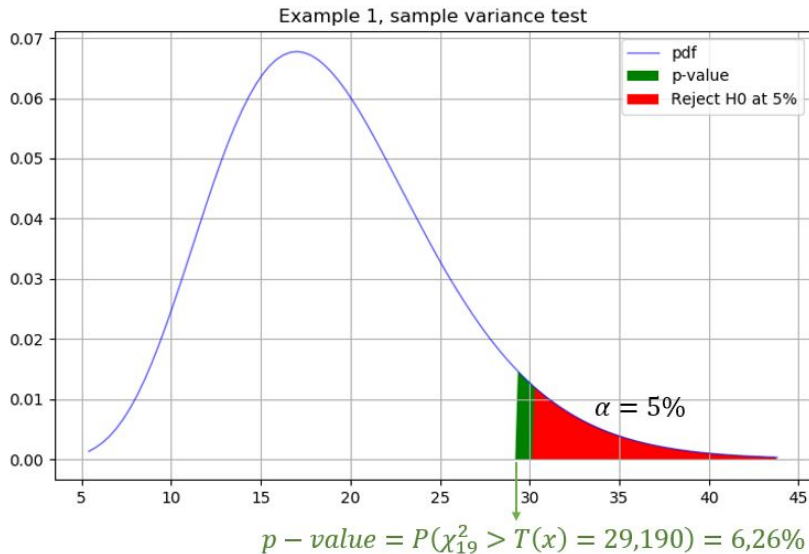
```
In [275]: pval = 1-sc.chi2.cdf(Tx,df=n-1)
...: print(pval)
...:
0.06291083863033275
```

p-value > 5%. We do not reject
 $H_0 : \sigma^2 = 20^2$ for $\alpha = 5\%$

P-value



P-value



Hypothesis testing, 2 populations

Comparison of 2 means

We consider two i.i.d 2 populations with the **same variance**:

$$\mathbf{X}_1 = \{X_{1,1}, \dots, X_{1,n}\} \sim \mathcal{N}(\mu_1, \sigma^2) \Rightarrow \bar{X}_1 \sim \mathcal{N}\left(\mu_1, \frac{\sigma^2}{n_1}\right)$$

$$\mathbf{X}_2 = \{X_{2,1}, \dots, X_{2,n}\} \sim \mathcal{N}(\mu_2, \sigma^2) \Rightarrow \bar{X}_2 \sim \mathcal{N}\left(\mu_2, \frac{\sigma^2}{n_2}\right)$$

We test if the 2 samples have the same means, or more generally :

$$H_0 : \mu_1 - \mu_2 = \delta$$

Against

$$a) H_1 : \mu_1 - \mu_2 > \delta$$

$$b) H_1 : \mu_1 - \mu_2 < \delta$$

$$c) H_1 : \mu_1 - \mu_2 \neq \delta$$

2 cases: σ known and unknown.

Comparison of 2 means

Case 1 σ known: If we remember the properties of normal r.v.

$$\bar{X}_1 - \bar{X}_2 \sim \mathcal{N}\left(\mu_1 - \mu_2, \frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}\right)$$

From equation (4), we use as statistics of test:

$$T(\mathbf{X}_1, \mathbf{X}_2) = \frac{(\bar{X}_1 - \bar{X}_2) - \delta}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim \mathcal{N}(0, 1) \quad (8)$$

We reject $H_0 : \mu_1 - \mu_2 = \delta$ at the level α if

- a) $H_1 : \mu_1 - \mu_2 > \delta \quad T(\mathbf{x}_1, \mathbf{x}_2) > z_{1-\alpha}$
- b) $H_1 : \mu_1 - \mu_2 < \delta \quad T(\mathbf{x}_1, \mathbf{x}_2) < z_{\alpha}$
- c) $H_1 : \mu_1 - \mu_2 \neq \delta \quad T(\mathbf{x}_1, \mathbf{x}_2) < z_{\alpha/2} \text{ or } T(\mathbf{x}_1, \mathbf{x}_2) > z_{1-\alpha/2}$

where e.g. z_{α} is the percentile of a $Z \sim \mathcal{N}(0, 1)$ ($P(Z \leq z_{\alpha}) = \alpha$)

Comparison of 2 means

Case 2 σ unknown: From equation (7), we use as statistics of test:

$$T(\mathbf{X}_1, \mathbf{X}_2) = \frac{(\bar{X}_1 - \bar{X}_2) - \delta}{S_{pool} \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{n_1+n_2-2} \quad (9)$$

We reject $H_0 : \mu_1 - \mu_2 = \delta$ at the level α if

- | | | | |
|----|-----------------------------------|---|----|
| a) | $H_1 : \mu_1 - \mu_2 > \delta$ | $T(\mathbf{x}_1, \mathbf{x}_2) > t_{n_1+n_2-2} \mathbf{1-\alpha}$ | |
| b) | $H_1 : \mu_1 - \mu_2 < \delta$ | $T(\mathbf{x}_1, \mathbf{x}_2) < t_{n_1+n_2-2} \alpha$ | |
| c) | $H_1 : \mu_1 - \mu_2 \neq \delta$ | $\begin{cases} T(\mathbf{x}_1, \mathbf{x}_2) < t_{n_1+n_2-2} \alpha/2 \\ T(\mathbf{x}_1, \mathbf{x}_2) > t_{n_1+n_2-2} \mathbf{1-\alpha/2} \end{cases}$ | or |

where e.g. $t_{n_1+n_2-2} \alpha$ is the α -percentile of a Student's T.

Comparison of 2 means

Example 1. `ComparisonMeanTest.py` Two machines fill bottles to a volume of 500 ml (normal distribution). The manufacturer believes that machines 1 and 2 fills different volumes. A sample of 2x20 bottles is taken and the volume inside is measured.

Volume	Volume
Mach. 1	Mach. 2
484.11	525.19
...	...
502.85	516.3
...	...
449.08	517.7
489.27	512.63

`scipy.stats.` $H_0 : \mu_1 - \mu_2 = 0$, $H_1 : \mu_1 - \mu_2 \neq 0$

```
In [314]: Xb1 = np.mean(X1)
...: Xb2 = np.mean(X2)
...: S1 = np.std(X1,ddof=1)
...: S2 = np.std(X2,ddof=1)
...: Spool = np.sqrt(((n-1)*S1**2+(n-1)*S2**2)/(n+n-2))
...: Tx = (Xb1-Xb2)/(Spool*np.sqrt(1/n+1/n))
...: # we compare it to percentiles of a t distribution
...: alpha = 0.05
...: t_l = sc.t.ppf(q=alpha/2,df=n+n-2)
...: t_u = sc.t.ppf(q=1-alpha/2,df=n+n-2)
...: #we see that Tx is in the 2.5% and 97.5% interval of the
Student's t
...: print([Tx,t_l,t_u])
[-4.013925858585345, -2.0243941645751367, 2.024394164575136]

In [315]: pval = 2*sc.t.cdf(-np.abs(Tx),df=n+n-2)
...: print(pval)
0.0002709539978837068
```

p-value=0.02%<5%, We **reject** H_0 for $\alpha = 5\%$.

Other command: `ttest_ind(X1,X2)`

Comparison of 2 variances

We consider two i.i.d 2 populations with **different variances**:

$$\mathbf{X}_1 = \{X_{1,1}, \dots, X_{1,n}\} \sim \mathcal{N}(\mu_1, \sigma_1^2)$$

$$\mathbf{X}_2 = \{X_{2,1}, \dots, X_{2,n}\} \sim \mathcal{N}(\mu_2, \sigma_2^2)$$

We test if the 2 samples have the same variances i.e.:

$$H_0 : \sigma_1 = \sigma_2$$

Against

$$a) H_1 : \sigma_1 \neq \sigma_2$$

$$b) H_1 : \sigma_1 > \sigma_2$$

$$c) H_1 : \sigma_1 < \sigma_2$$

Comparison of 2 variances

From equation (5), we use as statistics of test (Fisher's test):

$$T(\mathbf{X}_1, \mathbf{X}_2) = \frac{S_1^2}{S_2^2} \sim F_{n_1-1, n_2-1} \quad (10)$$

We reject $H_0 : \sigma_1^2 = \sigma_2^2$ at the level α if

$$\begin{array}{ll} \text{a) } H_1 : \sigma_1 \neq \sigma_2 & \begin{cases} T(\mathbf{x}_1, \mathbf{x}_2) < F_{n_1-1, n_2-1, \alpha/2} \\ T(\mathbf{x}_1, \mathbf{x}_2) > F_{n_1-1, n_2-1, 1-\alpha/2} \end{cases} \quad \text{or} \\ \text{b) } H_1 : \sigma_1 > \sigma_2 & T(\mathbf{x}_1, \mathbf{x}_2) > F_{n_1-1, n_2-1, 1-\alpha} \\ \text{c) } H_1 : \sigma_1 < \sigma_2 & T(\mathbf{x}_1, \mathbf{x}_2) < F_{n_1-1, n_2-1, \alpha} \end{array}$$

where e.g. $F_{n_1-1, n_2-1, \alpha}$ is the α -percentile of a Fisher.

Comparison of 2 variances

Example 1. `ComparisonVarianceTest.py` Two machines fill bottles with soft drink to a volume of 500 ml. The actual volume is known to follow a normal distribution. Do filled volumes by machine 1 have a bigger variance than those of machine 2?

Volume	Volume
Mach. 1	Mach. 2
484.11	525.19
...	...
502.85	516.3
...	...
449.08	517.7
489.27	512.63

Package `scipy.stats`. $H_0 : \sigma_1 = \sigma_2$, $H_1 : \sigma_1 > \sigma_2$

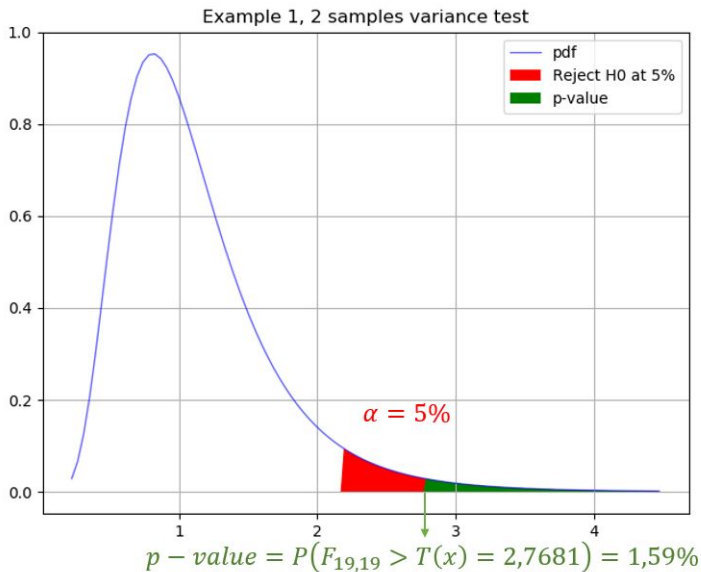
```
In [56]: S1      = np.std(X1,ddof=1)
...: S2      = np.std(X2,ddof=1)
...: Tx      = S1**2/S2**2
...: # we compare it to percentiles of a t distribution
...: alpha = 0.05
...: f_u    = sc.f.ppf(q=1-alpha,dfn=n-1, dfd=n-1)
...: #we see that Tx is in the 2.5% and 97.5% interval of the
Student's t
...: print([Tx,f_u])
[2.767818462651005, 2.168251601406261]

In [57]: pval = 1-sc.f.cdf(Tx,dfn=n-1 , dfd=n-1)
...: print(pval)
...:
0.01594817166131013
```

p-value 1.59% < 5%, We **reject** H_0 for $\alpha = 5\%$.

Other command: `Bartlett(X1, X2)` but it is a two-sided test, i.e. $H_1 : \sigma_1 \neq \sigma_2$!

Comparison of 2 variances



Linear regression

Linear regression

- ▶ We collect data to understand patterns and make predictions.
Examples:
 - ▶ Predicting house prices (responses) based on size and location (features).
 - ▶ Estimating a student's grade (responses) based on study hours (features).
 - ▶ Forecasting sales (responses) from advertising spend (features).
- ▶ When plotting data, we often observe a linear trend linking the responses to features.
- ▶ The linear regression assumes a straight-line relationship and provides statistical indicators about the significance of this assumption.

Linear regression

We observe n realizations y_i of Y_i that is related to d factors $(x_{i,1}, \dots, x_{i,d})^\top$ for $i = 1, \dots, n$. We postulate the following linear relation

$$Y_i = \beta_0 + \beta_1 x_{i,1} + \dots + \beta_d x_{i,d} + \epsilon_i \quad i = 1, \dots, n \quad (11)$$

where $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$. Matrix notations:

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix} \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_d \end{pmatrix} \quad \mathbf{X} = \begin{pmatrix} 1 & x_{1,1} & \dots & x_{1,d} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n,1} & \dots & x_{n,d} \end{pmatrix}$$

$\mathbf{Y}$ is a n vector, $\boldsymbol{\beta}$ is a $d + 1$ vector and $\mathbf{X}$ a $n \times (d + 1)$ matrix.

Linear regression

The n vector of noises is noted ϵ . Notice that realizations (observations) of $\mathbf{Y}$ (random vector) is a vector noted $\mathbf{y}$

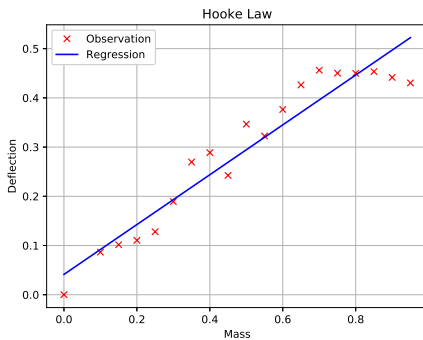
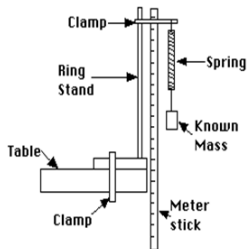
$$\epsilon = \begin{pmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{pmatrix} \quad \mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$$

Eq. (11) is then reformulated as

$$\mathbf{Y} = \mathbf{X}\beta + \epsilon$$

Linear regression

Example, Hooke's law experiments. A spring is loaded with a given weight M , and the resulting deflection D is measured. A linear relationship $D = \beta_1 M$ is expected. There are 19 records, mass in kilograms, and deflection in meters (see file [springs.csv](#)).



- ▶ The best $\hat{y}$ prediction of Y for a given value of $\mathbf{x} = (1, x_1, \dots, x_d)^\top$ is:

$$\hat{y} = \mathbb{E}(Y | \mathbf{x}) = \mathbf{x}^\top \boldsymbol{\beta}.$$

- ▶ However $\boldsymbol{\beta}$ and σ^2 are unknown. We denote by $\hat{\boldsymbol{\beta}}$ and $\hat{\sigma}^2$ their estimates.
- ▶ How do we estimate the parameters $\boldsymbol{\beta}$ and σ^2 based on a data sample $(\mathbf{x}_i, y_i)$ for $i = 1, \dots, n$?
- ▶ Methods of estimation: **Likelihood maximization**

Linear regression

We have $\theta = (\beta, \sigma)$ and conditionally to $\mathbf{x}_i$, the response is normal:

$$Y_i | \mathbf{x}_i \sim \mathcal{N}(\mathbf{x}_i^\top \beta; \sigma^2) \quad i = 1, \dots, n$$

The estimate of θ maximizes the log-likelihood function:

$$\begin{aligned} l(\theta) &= \sum_{i=1}^n \ln \left(\frac{1}{\sigma \sqrt{2\pi}} \exp \left(-\frac{1}{2} \frac{(y_i - \mathbf{x}_i^\top \beta)^2}{\sigma^2} \right) \right) \quad (12) \\ &= \sum_{i=1}^n \left(-\ln(\sigma \sqrt{2\pi}) - \frac{1}{2} \left(\frac{(y_i - \mathbf{x}_i^\top \beta)^2}{\sigma^2} \right) \right) \\ &\propto - \underbrace{\sum_{i=1}^n (y_i - \mathbf{x}_i^\top \beta)^2}_{(*)} \end{aligned}$$

Therefore, for a given $\sigma > 0$, the estimates of β minimizes (*).

Linear regression

Least Squares Minimization. The estimate $\hat{\beta}$ of β minimizes the sum of squared errors (**SSE**):

$$\begin{aligned}\hat{\beta} &= \arg_{\beta} \min (\mathbf{y} - \mathbf{X}\beta)^{\top} (\mathbf{y} - \mathbf{X}\beta) \\ &= \arg_{\beta} \min \sum_{i=1}^n (y_i - \mathbf{x}_i^{\top} \beta)^2\end{aligned}\tag{13}$$

which is:

$$\hat{\beta} = (\mathbf{X}^{\top} \mathbf{X})^{-1} \mathbf{X}^{\top} \mathbf{y}.\tag{14}$$

Proof. Eq. (13) is a direct consequence of (*). If we derive it w.r.t. β and cancel the derivative:

$$2\mathbf{X}^{\top} (\mathbf{y} - \mathbf{X}\beta) = 0\tag{15}$$

which admits Eq. (14) as solution.

Linear regression

The best prediction of Y_i for a vector of factor $\mathbf{x}_i$ is

$$\hat{y}_i = \mathbf{x}_i^\top \hat{\boldsymbol{\beta}} \quad i = 1, \dots, n$$

or under matrix form, if $\hat{\mathbf{y}} = (y_1, \dots, y_n)^\top$, $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$. By construction

$$\begin{aligned} \hat{\mathbf{y}} &= \underbrace{\mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top}_{\mathbf{H}} \mathbf{y} \\ &= \mathbf{H} \mathbf{y} \end{aligned}$$

where $\mathbf{H}$ is the **hat matrix**. This matrix is symmetric and idempotent, i.e. :

$$\mathbf{H} = \mathbf{H}^\top \quad \mathbf{H}\mathbf{H} = \mathbf{H}.$$

If the assumption of normality does not hold, the LSM estimators are not the ML estimators **but we still can use the Least Squares Minimization**.

Simple linear regression

Simple regression: Only one explanatory factor (simple regression, $d = 1$):

$$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

Then

$$\{\hat{\beta}_0, \hat{\beta}_1\} = \arg_{\beta_0, \beta_1} \min \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2 \quad , \quad (16)$$

Deriving Eq. (16) w.r.t. β_0 and β_1 and cancelling the differentials leads to

$$\begin{cases} n\beta_0 + \beta_1 \sum_{i=1}^n x_i &= \sum_{i=1}^n y_i \\ \beta_0 \sum_{i=1}^n x_i + \beta_1 \sum_{i=1}^n x_i^2 &= \sum_{i=1}^n x_i y_i \end{cases}$$

Simple linear regression

Simple regression: The intercept $\hat{\beta}_0$ and the slope, $\hat{\beta}_1$, are:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Example, Hooke's law experiments. See [regressionSpring.py](#). Either direct calculation or `linregress(...)`. We find $\hat{\beta}_1 = 0.506$ and $\hat{\beta}_0 = 0.041$

```
In [75]: X = datn[:,0] ;   Y = datn[:,1]
...: Xb=np.mean(X) ;   Yb= np.mean(Y)
...: b1=np.sum((X-Xb)*(Y-Yb))/np.sum((X-Xb)**2)
...: b0=Yb-b1*Xb
...: # alternative scipy linregress(.)
...: slope, intercept, r_value, p_value, std_err =
sc.linregress(X,Y)
...: print(np.round([b0,intercept,b1,slope],3))
...:
[0.041 0.041 0.506 0.506]
```

Linear regression

- ▶ How do we estimate the quality of the model? We split the variance of observations into:
 - ▶ A variance explained by the model
 - ▶ A residual variance
- ▶ The higher is the variance explained by the model, the better is the **goodness of fit**.

Let us note $\hat{y}_i = \mathbf{x}_i^\top \hat{\boldsymbol{\beta}}$ then

$$\underbrace{\sum_{i=1}^n (y_i - \bar{y})^2}_{SS \text{ Total}} = \underbrace{\sum_{i=1}^n (y_i - \hat{y}_i)^2}_{SS \text{ Error}} + \underbrace{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}_{SS \text{ Regression}}$$

Linear regression

Proof. We have that $(y_i - \bar{y}) = (y_i - \hat{y}_i) + (\hat{y}_i - \bar{y})$ and according to Eq. (15):

$$\begin{aligned}\sum_{i=1}^n (y_i - \hat{y}_i) (\hat{y}_i - \bar{y}) &= \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \hat{\boldsymbol{\beta}}) (\mathbf{x}_i^\top \hat{\boldsymbol{\beta}} - \bar{y}) \\ &= (\hat{\beta}_0 - \bar{y}) \underbrace{\sum_{i=1}^n (y_i - \mathbf{x}_i^\top \hat{\boldsymbol{\beta}})}_{\frac{1}{2} \frac{\partial SSE}{\partial \hat{\beta}_0} = 0} + \sum_{j=1}^k \hat{\beta}_j \underbrace{\sum_{i=1}^n \mathbf{x}_{i,j} (y_i - \mathbf{x}_i^\top \hat{\boldsymbol{\beta}})}_{\frac{1}{2} \frac{\partial SSE}{\partial \hat{\beta}_j} = 0} = 0\end{aligned}$$

The $R^2 \in [0, 1]$ is the proportion of the variance explained by the model:

$$R^2 = \frac{SSR}{SST} \Leftrightarrow 1 - R^2 = \frac{SSE}{SST}.$$

The closer to unity, the better is the model.

Linear regression

Example, Hooke's law experiments. See `regressionSpring.py`. R^2 computed by `linregress(...)`, $R^2 = 99.88\%$! Excellent fit.

Remark: Matrix notations of the SST, SSR, SSE. If I_n is the $n \times n$ identity matrix and J_n is the $n \times n$ matrix of ones:

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2 = \mathbf{y}^\top \left(I_n - \frac{1}{n} J_n \right) \mathbf{y}$$

$$SSR = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 = \mathbf{y}^\top \left(H - \frac{1}{n} J_n \right) \mathbf{y}$$

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \mathbf{y}^\top (I_n - H) \mathbf{y}$$

Properties of regression coefficients

Linear regression

$\hat{\beta}$ is an **unbiased Gaussian estimator** of β (i.e. $\mathbb{E}(\hat{\beta}) = \beta$):

$$\hat{\beta} \sim \mathcal{N}\left(\beta, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}\right)$$

Remark: here $\hat{\beta}$ is a multivariate normal, see appendix.

Proof 1) The estimator is a random variable $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$ where $\mathbf{Y} = (Y_1, \dots, Y_n)^\top = \mathbf{X}\beta + \epsilon$. Since a linear combination of Normal r.v.'s is Normal, $\hat{\beta}$ is normal. 2) Since $\mathbb{E}(\mathbf{Y}) = \mathbf{X}\beta$, the estimator is unbiased:

$$\begin{aligned}\mathbb{E}(\hat{\beta}) &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbb{E}(\mathbf{Y}) \\ &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} \beta = \beta.\end{aligned}$$

Linear regression

2) Using standard linear algebra, we infer that:

$$\begin{aligned}\mathbb{V}(\hat{\beta}) &= \mathbb{V}\left(\left(\mathbf{X}^\top \mathbf{X}\right)^{-1} \mathbf{X}^\top \mathbf{Y}\right) \\ &= \left(\mathbf{X}^\top \mathbf{X}\right)^{-1} \mathbf{X}^\top \mathbb{V}(\mathbf{Y}) \mathbf{X} \left(\left(\mathbf{X}^\top \mathbf{X}\right)^{-1}\right)^\top = \sigma^2 \left(\mathbf{X}^\top \mathbf{X}\right)^{-1}\end{aligned}$$

An **unbiased** estimator of σ^2 is

$$\hat{\sigma}^2 = \frac{1}{n - (d + 1)} \underbrace{\left(\mathbf{Y} - \mathbf{X}\hat{\beta}\right)^\top \left(\mathbf{Y} - \mathbf{X}\hat{\beta}\right)}_{SSE}$$

and

$$(n - (d + 1)) \frac{\hat{\sigma}^2}{\sigma^2} = \frac{SSE}{\sigma^2} \sim \chi^2_{n-(d+1)}$$

is a **chi-square random variable** with $n - (d + 1)$ degree of freedoms.

Linear regression (for information)

Proof: The trace of $\mathbf{H}$ is

$$\begin{aligned} \text{tr}(\mathbf{H}) &= \text{tr}\left(\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top\right) = \text{tr}\left((\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}\right) \\ &= \text{tr}(\mathbf{I}_{d+1}) = d + 1. \end{aligned}$$

We next use the properties that eigenvalues of idempotent matrix ($\mathbf{H}^2 = \mathbf{H}$) must be equal to zero or one. To prove this, let ϕ the normalized eigenvector of $\mathbf{H}$ with eigenvalue λ : $\mathbf{H}\phi = \lambda\phi$ then

$$\underbrace{\mathbf{H}\mathbf{H}}_{=\mathbf{H}}\phi = \lambda \underbrace{\mathbf{H}\phi}_{\lambda\phi} \Rightarrow \lambda\phi = \lambda^2\phi$$

and $\lambda = 0$ or 1 . The matrix $\mathbf{I}_n - \mathbf{H}$ is also idempotent because:

$$(\mathbf{I}_n - \mathbf{H})(\mathbf{I}_n - \mathbf{H}) = \mathbf{I}_n - \mathbf{I}_n\mathbf{H} - \mathbf{H}\mathbf{I}_n + \mathbf{H}^2 = \mathbf{I}_n - \mathbf{H}$$

Linear regression (for information)

Proof (cont'd) Its trace is also the sum of eigenvalues:

$$\text{tr}(\mathbf{I}_n - \mathbf{H}) = \text{tr}(\mathbf{I}_n) - \text{tr}(\mathbf{H}) = n - (d + 1)$$

then $\mathbf{I}_n - \mathbf{H}$ has $n - (d + 1)$ eigenvalues equal to 1 and $(d + 1)$ equal to zero. The spectral decomposition of $\mathbf{I}_n - \mathbf{H}$ is $\mathbf{A}\mathbf{D}\mathbf{A}^\top$ where $\mathbf{A}\mathbf{A}^\top = \mathbf{I}_n$ because $\mathbf{I}_n - \mathbf{H}$ is symmetric and

$$\mathbf{D} = \begin{pmatrix} \mathbf{I}_{n-d-1} & \mathbf{0}_{(n-d-1) \times (d+1)} \\ \mathbf{0}_{(d+1) \times (n-d-1)} & \mathbf{0}_{(d+1) \times (d+1)} \end{pmatrix}$$

Since $\mathbf{D} = \mathbf{D}^\top$ and $\mathbf{D} = \mathbf{D}\mathbf{D}$:

$$\text{SSE} = \mathbf{Y}^\top \mathbf{A}\mathbf{D}^\top \mathbf{A}^\top \mathbf{Y} = (\mathbf{Y}^\top \mathbf{A}\mathbf{D}) (\mathbf{D}^\top \mathbf{A}^\top \mathbf{Y})$$

Given that $\mathbf{Y} \sim \mathcal{N}(\mathbf{X}\beta, \sigma^2 \mathbf{I}_n)$, using standard normal theory:

$$\mathbf{D}^\top \mathbf{A}^\top \mathbf{Y} \sim \mathcal{N}(\mathbf{D}^\top \mathbf{A}^\top \mathbf{X}\beta, \sigma^2 \mathbf{D}^\top \mathbf{A}^\top \mathbf{A}\mathbf{D}) \sim \mathcal{N}(\mathbf{D}^\top \mathbf{A}^\top \mathbf{X}\beta, \sigma^2 \mathbf{D})$$

Linear regression (for information)

Proof (cont'd) showing that components of $\mathbf{D}^\top \mathbf{A}^\top \mathbf{Y}$ are independent. Furthermore, $\left(\mathbf{D}^\top \mathbf{A}^\top \mathbf{Y}\right)_i \sim \mathcal{N}(0, \sigma^2)$ for $i = 1, \dots, n - d - 1$ and null otherwise. Then $\left(\mathbf{Y}^\top \mathbf{A} \mathbf{D} \mathbf{A}^\top \mathbf{Y}\right) / \sigma^2$ is χ_{n-d-1}^2 with expectation equal to $n - d - 1$:

$$\mathbb{E} \left(\left(\mathbf{Y}^\top \mathbf{A} \mathbf{D} \mathbf{A}^\top \mathbf{Y} \right) / \sigma^2 \right) = n - d - 1$$

Therefore the estimator is unbiased:

$$\mathbb{E}(\hat{\sigma}^2) = \mathbb{E} \left(\frac{\left(\mathbf{Y}^\top \mathbf{A} \mathbf{D} \mathbf{A}^\top \mathbf{Y} \right)}{n - d - 1} \right) = \sigma^2.$$

end

Simple linear regression

$\hat{\beta}_0$ and $\hat{\beta}_1$ are unbiased estimators i.e. $\mathbb{E}(\hat{\beta}_0) = \beta_0$ and $\mathbb{E}(\hat{\beta}_1) = \beta_1$. If $\bar{x}^2 = \frac{1}{n} \sum_{i=1}^n x_i^2$, their variances are

$$\mathbb{V}(\hat{\beta}_0) = \frac{\sigma^2 \bar{x}^2}{S_{xx}} \quad \mathbb{V}(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}} \quad \mathbb{C}(\hat{\beta}_0, \hat{\beta}_1) = \frac{-\sigma^2 \bar{x}}{S_{xx}}$$

where $S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$ (but σ^2 is unknown...)

Since $\epsilon \sim \mathcal{N}(0, \sigma^2)$ then

$$\hat{\beta}_0 \sim \mathcal{N}\left(\beta_0, \frac{\sigma^2 \bar{x}^2}{S_{xx}}\right) \quad \hat{\beta}_1 \sim \mathcal{N}\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right)$$

$$(n-2) \frac{\hat{\sigma}^2}{\sigma^2} = \frac{SSE}{\sigma^2} \sim \chi_{n-2}^2$$

Simple Linear regression

In practice σ^2 is unknown and we replace it by $\hat{\sigma}^2$: $\mathbb{V}(\hat{\beta}_0) = \frac{\hat{\sigma}^2 \overline{x^2}}{S_{xx}}$ and $\mathbb{V}(\hat{\beta}_1) = \frac{\hat{\sigma}^2}{S_{xx}}$: consequence $\hat{\beta}_0$ and $\hat{\beta}_1$ become Student's T!

Example, Hooke's law experiments. See [regressionSpring.py](#). Best package for linear regression: "[statsmodels](#)". Function `OLS(Y,X).fit`

- ▶ According to theory, β_0 should be zero... but we find $\hat{\beta}_0 = 0.04$. However the standard deviation is high $\sqrt{\frac{\hat{\sigma}^2 \overline{x^2}}{S_{xx}}} = 0.02$ compared to $\hat{\beta}_0$! $\hat{\beta}_0$ is then very inaccurate.
- ▶ The standard deviation of $\hat{\beta}_1$ is $\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}} = 0.02$, low compared to $\hat{\beta}_1 = 0.5064$. $\hat{\beta}_0$ is then reliable.

	coef	std err
const	0.0413	0.021
x1	0.5064	0.036

Test of the significance of linear regression

From properties of S^2 , if $\beta_1 = \dots = \beta_d = 0$ (H_0), the normalized **SST** is a chi-square r.v. with $n - 1$ d.f.

$$\frac{\text{SST}}{\sigma^2} = \frac{1}{\sigma^2} \sum_{i=1}^n (Y_i - \bar{Y})^2 \sim \chi_{n-1}^2$$

From this section, the normalized **SSE** a chi-square r.v. with $n - (d + 1)$ d.f.

$$\frac{\text{SSE}}{\sigma^2} = \frac{1}{\sigma^2} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 \sim \chi_{n-(d+1)}^2$$

Test of the significance of linear regression

Since

$$SST = SSE + SSR$$

and as a χ_n^2 r.v. is a sum of n r.v. :

$$\frac{SSR}{\sigma^2} = \frac{1}{\sigma^2} \sum_{i=1}^n \left(\hat{Y}_i - \bar{Y} \right)^2 \sim \chi_d^2$$

Therefore, if $\beta_1 = \dots = \beta_d = 0$, the next ratio is a Fisher r.v.:

$$F^* = \frac{SSR / d}{SSE / (n - d - 1)} \sim F_{d, n-(d+1)}$$

Test of the significance of linear regression

If F^* (to interpret as **explained variance on unexplained variance**) is “too high” then the assumption H_0 must be rejected and we admit a linear relation.

Significance test of the regression:

$$H_0 : \beta_1 = \dots = \beta_d = 0$$

$$H_1 : \beta_j \neq 0 \text{ for some } j \in \{1, \dots, d\}$$

with the statistics of test F^* :

$$F^* = \frac{MSR}{MSE} = \frac{SSR / d}{SSE / (n - d - 1)} \sim F_{d, n-(d+1)}$$

Reject H_0 at a confidence level α (e.g. 5%)

- ▶ If $F^* > F_{d, n-(d+1), 1-\alpha}$
- ▶ if the p-value $p_{val} = P(F^* < F_{d, n-(d+1)})$ is lower than α

Significance of a simple linear regression

Example: Hooke's law experiments. In a simple framework,
 $Y_i = \beta_0 + \beta_1 x_i + \epsilon_i$

	d.f.	Value	Mean of ...	F^*	p-value 5%
SSR	1 (d)	0.378	0.378	193.68	1.e-10
SSE	17($n - 2$)	0.033	0.0019		
SST	18($n - 1$)	0.411			

For $\alpha = 5\%$, $F_{1,17,1-\alpha} = 4.45$. Since $F^* > F_{1,17,0.95}$ and $p_{val} < 5\%$, we reject H_0 .

See also [regressionSpring.py](#). Package statsmodels,

```
F-statistic:          193.7
Prob (F-statistic):    1.01e-10
```

Tests of regression coefficients

Let $c_{j,j}$ be the j^{th} diagonal element of $(\mathbf{X}^\top \mathbf{X})^{-1}$. Estimators $\hat{\boldsymbol{\beta}} = (\hat{\beta}_0, \dots, \hat{\beta}_d)^\top$ of linear regression coefficients are Student's T r.v.:

$$\frac{\hat{\beta}_j - \beta_j}{\hat{\sigma} \sqrt{c_{jj}}} \sim t_{n-(d+1)}$$

Proof. $\hat{\boldsymbol{\beta}}$ is a multivariate normal $\mathcal{N}(\boldsymbol{\beta}, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1})$ then

$$\frac{\hat{\beta}_j - \beta_j}{\sqrt{\sigma^2 c_{jj}}} \sim \mathcal{N}(0, 1)$$

The estimator $\hat{\sigma}^2$ of variance σ^2 is $(n - (d + 1)) \frac{\hat{\sigma}^2}{\sigma^2} \sim \chi_{n-(d+1)}^2$. By definition of the Student's T, we conclude that

$$\frac{\hat{\beta}_j - \beta_j}{\hat{\sigma} \sqrt{c_{jj}}} = \frac{\hat{\beta}_j - \beta_j}{\sqrt{\sigma^2 c_{jj}}} \bigg/ \sqrt{\frac{n - (d + 1)}{n - (d + 1)} \frac{\hat{\sigma}^2}{\sigma^2}} \sim t_{n-(d+1)}$$

Tests of regression coefficients

We use this result to test the significance of each β_j :

$$H_0 : \beta_j = \beta_{j,0}$$

against 3 alternatives:

$$a) H_1 : \beta_j > \beta_{j,0}$$

$$b) H_1 : \beta_j < \beta_{j,0}$$

$$c) H_1 : \beta_j \neq \beta_{j,0}$$

with the statistics of test $T_j^* = \frac{\hat{\beta}_j - \beta_{j,0}}{\hat{\sigma} \sqrt{c_{jj}}} \sim t_{n-(d+1)}$. We reject H_0 at the level α if T_j^*

$$a) H_1 : \beta_j > \beta_{j,0} \quad T_j^* > t_{n-d-1} \text{ } 1-\alpha$$

$$b) H_1 : \beta_j < \beta_{j,0} \quad T_j^* < t_{n-d-1} \text{ } \alpha$$

$$c) H_1 : \beta_j \neq \beta_{j,0} \quad T_j^* < t_{n-d-1} \text{ } \alpha/2 \text{ or } T_j^* > t_{n-d-1} \text{ } 1-\alpha/2$$

Tests of regression coefficients

A Confidence interval for β_j at level $1 - \alpha$ (e.g. $\alpha=5\%$) is an interval $[\beta_{j,L}, \beta_{j,U}]$ such that β_j is in this interval with a probability $1 - \alpha$.

Since $\frac{\hat{\beta}_j - \beta_j}{\hat{\sigma} \sqrt{c_{jj}}} \sim t_{n-(d+1)}$ we infer that

$$P \left(t_{n-d-1} \alpha/2 \leq \frac{\hat{\beta}_j - \beta_j}{\hat{\sigma} \sqrt{c_{jj}}} \leq t_{n-d-1} 1-\alpha/2 \right) = 1 - \alpha$$

The $1 - \alpha$ confidence interval is then

$$\left[\hat{\beta}_j - t_{n-d-1} \alpha/2 \hat{\sigma} \sqrt{c_{jj}}; \hat{\beta}_j + t_{n-d-1} 1-\alpha/2 \hat{\sigma} \sqrt{c_{jj}} \right]$$

Or (the Student's T is symmetric: $-t_{n-1} \alpha/2 = t_{n-1} 1-\alpha/2$)

$$\left[\hat{\beta}_j - t_{n-d-1} 1-\alpha/2 \hat{\sigma} \sqrt{c_{jj}}; \hat{\beta}_j + t_{n-d-1} \alpha/2 \hat{\sigma} \sqrt{c_{jj}} \right]$$

Test of β : simple linear regression

When one explanatory factor (simple regression, $d = 1$),
 $Y_i = \beta_0 - \beta_1 x_i + \epsilon_i$, the coefficient $c_{0,0}$ and $c_{1,1}$ are easy to calculate:

$$c_{0,0} = \frac{\overline{x^2}}{S_{xx}} \quad c_{1,1} = \frac{1}{S_{xx}}$$

where $\overline{x^2} = \frac{1}{n} \sum_{i=1}^n x_i^2$, and $S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$. We test

$$H_0 : \beta_1 = \beta_{1,0}$$

$$H_1 : \beta_1 \neq \beta_{1,0}$$

with the statistics of test $T_1^* = \frac{\hat{\beta}_1 - \beta_{1,0}}{\hat{\sigma} \sqrt{S_{xx}^{-1}}} \sim t_{n-2}$. The confidence interval is

$$\beta_1 \in \left[\hat{\beta}_1 - t_{n-d-1, 1-\alpha/2} \hat{\sigma} \sqrt{S_{xx}^{-1}}; \hat{\beta}_1 + t_{n-d-1, 1-\alpha/2} \hat{\sigma} \sqrt{S_{xx}^{-1}} \right]$$

Test of β : simple linear regression

b) We test

$$H_0 : \beta_0 = \beta_{0,0}$$

$$H_1 : \beta_0 \neq \beta_{0,0}$$

with the statistics of test $T_0^* = \frac{\hat{\beta}_0 - \beta_{0,0}}{\hat{\sigma} \sqrt{\bar{x}^2 S_{xx}^{-1}}} \sim t_{n-2}$. The confidence interval is

$$\beta_0 \in \left[\hat{\beta}_0 - t_{n-d-1, 1-\alpha/2} \hat{\sigma} \sqrt{\bar{x}^2 S_{xx}^{-1}}; \hat{\beta}_0 + t_{n-d-1, 1-\alpha/2} \hat{\sigma} \sqrt{\bar{x}^2 S_{xx}^{-1}} \right]$$

Example: Hooke's law experiments. See file `regressionSpring.py`. The `OLS(.)` command from `stats.model` compute statistics, p-value and 95% confidence intervals.

Tests of regression coefficients

	coef	std err	t	P> t	[0.025	0.975]
const	0.0413	0.021	1.991	0.063	-0.002	0.085
x1	0.5064	0.036	13.917	0.000	0.430	0.583

For $H_0 : \beta_1 = 0$, $H_1 : \beta_1 \neq 0$, the statistics is $T_1^* = 13.917$ and the p-value is $p = 2P(t_{n-2} > |T_1^*|) = 0$. Strong rejection of H_0 . The 95% confidence interval is:

$$\beta_1 \in [0.430 ; 0.583]$$

For $H_0 : \beta_0 = 0$, $H_1 : \beta_0 \neq 0$ the statistics is $T_0^* = 1.991$ and the p-value is $p = 2P(t_{n-2} > |T_0^*|) = 6.3\%$. Acceptation of H_0 for $\alpha = 5\%$. Notice that for $\alpha > 6.3\%$ we reject H_0 . The 95% confidence interval is:

$$\beta_0 \in [-0.002 ; 0.085]$$

Tests of regression coefficients

However, you can easily compute the statistics T_1^* and T_0^* p-values and confidence intervals directly with scipy and numpy:

```
#you can calculate the t stat yourself
Sxx  = sum((X-Xb)**2)           #Sxx
X2b  = np.mean(X**2)           #mean of X^2
Yhat = results.predict(Xm)      #prediction
sghat = np.sqrt(sum((Y-Yhat)**2)/(n-2)) #estimate of sigma
# Test 1: beta1 =0 v.s. beta1<>0
T1    = b1/(sghat*np.sqrt(Sxx**-1)) #test statistics
pval1 = 2*(1-sc.t.cdf(abs(T1),df=n-2)) #p-value
#confidence interval for beta1
CI1    = b1+[-sghat*np.sqrt(Sxx**-1)*sc.t.ppf(q=0.975,df=n-2), \
             +sghat*np.sqrt(Sxx**-1)*sc.t.ppf(q=0.975,df=n-2) ]
#very low pvalue , we reject H0 : b1=0 at 5%
# Test 2: beta0 =0 v.s. beta0<>0
T0    = b0/(sghat*np.sqrt(X2b*Sxx**-1)) #test statistics
pval0 = 2*(1-sc.t.cdf(abs(T0),df=n-2)) #p-value
#confidence interval for beta0
CI0    = b0+[-sghat*np.sqrt(X2b*Sxx**-1)*sc.t.ppf(q=0.975,df=n-2), \
             +sghat*np.sqrt(X2b*Sxx**-1)*sc.t.ppf(q=0.975,df=n-2) ]
```

Prediction interval, simple regression

For a model $Y = \beta_0 + \beta_1 X + \epsilon$, the prediction for an unobserved value $X = x_0$ is

$$\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0 .$$

The **prediction interval** for Y_0 at level α is provided by

$$\left[\hat{y}_0 - S_{pred} t_{n-2; 1-\frac{\alpha}{2}} ; \hat{y}_0 + S_{pred} t_{n-2; 1-\frac{\alpha}{2}} \right] ,$$

where

$$S_{pred}^2 = \hat{\sigma}^2 \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)$$

This result is a direct consequence of previous propositions. A detailed proof is proposed in the last session of exercise.

Analysis of Variance (ANOVA)

1 factor ANOVA

Example : we want to test the resistance of a ceramic produced in three different factories: is it the same on average?

Data : 10 measurements per factory

Obs.	Factory 1	Factory 2	Factory 3
1	1495	1486	1572
2	1574	1610	1647
3	1401	1538	1615
4	1456	1515	1475
5	1418	1501	1521
6	1404	1522	1540
7	1517	1532	1544
8	1491	1617	1556
9	1501	1524	1659
10	1553	1669	1527
Mean	1481	1551,4	1565,6
S2	3638,7	3542,3	3414,7



1 factor ANOVA

If μ_1 , μ_2 and μ_3 are respectively the mean resistance measured in each factory, we test (assumption normal dis. and same variance):

$$\begin{cases} H_0 : \mu_1 = \mu_2 = \mu_3 \\ H_1 : \exists i \neq j, \mu_i \neq \mu_j \end{cases} \quad (17)$$

We can think to do pairwise comparisons... but

- ▶ This is tedious when we have more than 3 sub-samples,
- ▶ The risk of type 1 error increases (rejecting H_0 that is true).

Alternative : test of significance of a **categorical** linear regression.

1 factor ANOVA

In many statistical applications, we study the influence of categorical factors on the behaviour of a variable of interest (the response) by an experiment.

Example of ceramic production : does the production site impact the resistance of the produced material?

The response, Y , is the resistance of ceramic and the factor is a categorical variable “factory”, with 3 levels.

To reformulate this with a categorical regression we consider **2 binary** (or “dummy”) variables

$$X_1 = \begin{cases} 1 & \text{Fact. 1} \\ 0 & \text{else} \end{cases}, X_2 = \begin{cases} 1 & \text{Fact. 2} \\ 0 & \text{else} \end{cases}$$

`pandas.get_dummies(data, drop_first=False, dtype=None)`

1 factor ANOVA

“Dummified” dataset : $n = 30$ and $d = 2$,

i	y_i	$x_{i,1}$	$x_{i,2}$
1	1495	1	0
$\vdots$	$\vdots$	$\vdots$	$\vdots$
10	1553	1	0
11	1486	0	1
$\vdots$	$\vdots$	$\vdots$	$\vdots$
20	1669	0	1
21	1572	0	0
$\vdots$	$\vdots$	$\vdots$	$\vdots$
30	1527	0	0

Warning : do not introduce a third binary variables for the third factory! In this case, the matrix $\mathbf{X}$ is ill-posed !

1 factor ANOVA

Categorical linear regression : let $\epsilon \sim N(0, \sigma^2)$ then

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$$

Testing eq. (17) is equivalent to test

$$\begin{cases} H_0 : \beta_1 = \beta_2 = 0 \\ H_1 : \exists i, \beta_i \neq 0 \end{cases} \quad (18)$$

We recognize the global F significance test (with $n = 30$, $d = 2$)!

Underlying assumptions : normality, same variance of sub-samples, independence

1 factor ANOVA

	d.f.	Value	Mean of ...	F^*	p-value 5%
SSR	2 (d)	41050	20525	5.81	0.00796
SSE	27($n - 3$)	95361	3532		
SST	29($n - 1$)	136411			

Conclusion: we reject the hypothesis of identical average hardness!

Remark: the assumption of equality of variances can be checked with the Bartlett's test (extension of the bivariate variance test).

Auto-regressive models

Introduction to time-series

The objective of time series analysis is to provide models offering a plausible description for a sample of data indexed by time.

In the industry, it is used for planning production/demand e.g. of electricity. In finance, it is used for explaining the evolution of asset returns. In actuarial sciences, it is used for forecasting the number of claims caused by natural phenomenons.

A **time-series** is a vector indexed by time: $\mathbf{x} = (x_t)_{t=1,2,\dots,T}$, realizations of r.v. $(X_t)_{t=1,2,\dots,T}$. If the pdf of X_t is $f_t(x)$, the mean function μ_t is :

$$\mu_t = \mathbb{E}(X_t) = \int_{-\infty}^{+\infty} x f_t(x) dx ,$$

Introduction to time-series

The **autocovariance** function, $\gamma(t, s)$, is defined as the covariance between X_t and X_s :

$$\gamma(t, s) = \mathbb{E}[(X_t - \mu_t)(X_s - \mu_s)] \quad \forall t, s \in \mathbb{N}.$$

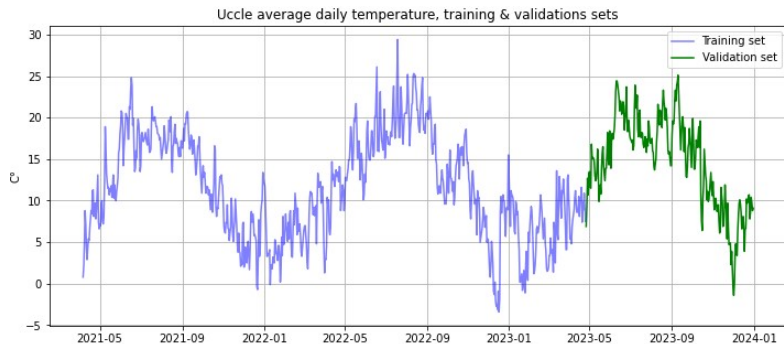
Smooth time series show a high autocovariances even when t and s are far apart. Highly volatile series have an autocovariance function quickly converging to zero when t moves away from s .

The **autocorrelation** function, $\rho(t, s)$, is:

$$\rho(t, s) = \frac{\gamma(t, s)}{\sqrt{\gamma(t, t)\gamma(s, s)}} \quad \forall t, s \in \mathbb{N}.$$

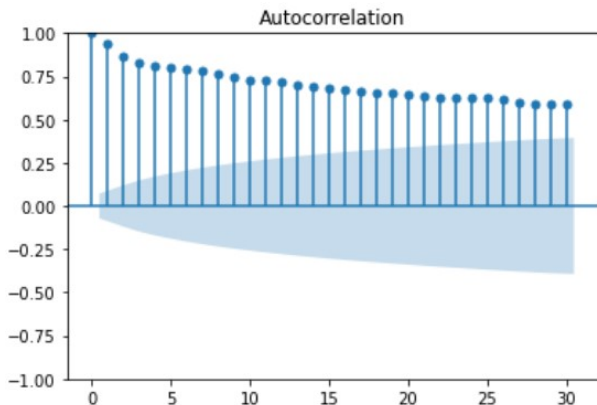
Introduction to time-series

Example: time series of average temperatures in Uccle from 4/6/21 to 31/12/23. We will fit a model on a “training set” (4/6/21 to 25/4/23) and test it on a validation sample (26/4/23 to 31/12/24)



Introduction to time-series

Example (cont'd) : Autocorrelation of temperatures (training set).
We observe a strong correlation between short-term values.



Autoregressive process

AR(p) : an **autoregressive process** of order p , $AR(p)$ is

$$\begin{aligned} X_t &= \beta_1 x_{t-1} + \beta_2 x_{t-2} + \dots + \beta_p x_{t-p} + \epsilon_t, \\ &= \boldsymbol{\beta}^\top \mathbf{x}_{t-1:t-p} + \epsilon_t, \end{aligned} \quad (19)$$

where ϵ_t is a Gaussian process with a null mean, $\mathbb{E}(\epsilon_t) = 0$ and variance, σ^2 .

By construction, the best prediction of X_t , knowing $\mathbf{x}_{t-1:t-p}$ is equal to:

$$\mathbb{E}(X_t | \mathbf{x}_{t-1:t-p}) = \boldsymbol{\beta}^\top \mathbf{x}_{t-1:t-p},$$

The variance of X_t , knowing $\mathbf{x}_{t-1:t-p}$ is equal to

$$\mathbb{V}(X_t | \mathbf{x}_{t-1:t-p}) = \sigma^2.$$

Autoregression

If we have n observations, this model is estimated as a standard linear regression model with a response vector $\mathbf{y}$ and matrix of covariates $\mathbf{X}$:

$$\mathbf{y} = \begin{pmatrix} x_{p+1} \\ \vdots \\ x_n \end{pmatrix} \quad \mathbf{X} = \begin{pmatrix} x_p & \dots & x_1 \\ \vdots & \ddots & \vdots \\ x_{n-1} & \dots & x_{n-p} \end{pmatrix}.$$

The estimator of β is obtained by log-likelihood maximization:

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

The estimate of $X_t | \mathbf{x}_{t-1:t-p}$ is then

$$\hat{x}_t = \hat{\beta}^\top \mathbf{x}_{t-1:t-p}$$

whereas the variance of predictions is estimated by

$$\hat{\sigma}^2 = \frac{1}{(T-p)-1} \sum_{t=p+1}^T (x_t - \hat{x}_t)^2.$$

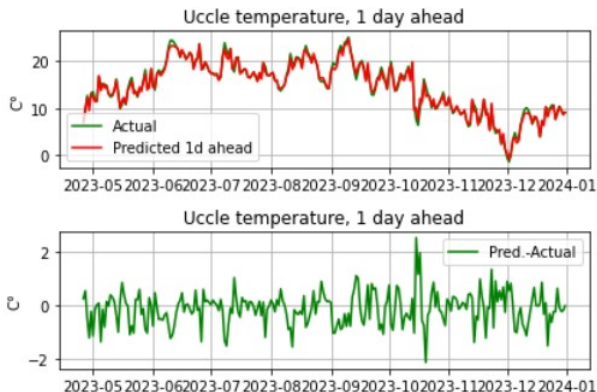
Autoregression

Example (cont'd) : Instead of using `statsmodels.ols(...)`, you can use `statsmodels.AutoReg(...)` but be careful when entering optional parameters. Here the autoregression with 5 lags of temperatures (with `trend='n'`), training set:

```
AutoReg Model Results
=====
Dep. Variable:                y      No. Observations:                750
Model:                        AutoReg(5)  Log Likelihood                -1582.848
Method:                      Conditional MLE  S.D. of innovations                2.025
Date:                        Fri, 04 Oct 2024  AIC                3177.697
Time:                        13:05:09      BIC                3205.377
Sample:                      5      HQIC                3188.366
                                750
=====
              coef      std err          z      P>|z|      [0.025      0.975]
-----
y.L1          1.0032        0.036     27.709      0.000        0.932        1.074
y.L2         -0.2949        0.051     -5.731      0.000       -0.396       -0.194
y.L3          0.1870        0.052      3.583      0.000        0.085        0.289
y.L4         -0.0559        0.052     -1.085      0.278       -0.157        0.045
y.L5          0.1528        0.036      4.231      0.000        0.082        0.224
```

Autoregression

Example (cont'd) : On the validation set (26/4/23 to 31/12/24), we compare forecast $\hat{x}_t$ and observed values:



Autoregression

We can also use the autoregression to forecast the process k periods ahead. Using nested expectations, we can show that an estimate $\hat{x}_{t+k}$ of $X_{t+k} | \mathbf{x}_{t-1:t-p}$ is equal to

$$\hat{x}_{t+k} = \hat{\beta}^\top \hat{\mathbf{x}}_{(t+k)-1:(t+k)-p}.$$

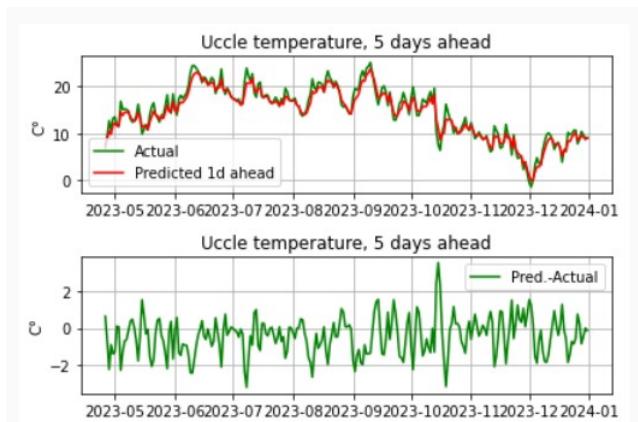
For instance if $p = 3$, conditionally to $\mathbf{x}_{t-1:t-3}$, we have the following forecasts

$$\begin{aligned}\hat{x}_t &= \hat{\beta}_1 x_{t-1} + \hat{\beta}_2 x_{t-2} + \hat{\beta}_3 x_{t-3} \\ \hat{x}_{t+1} &= \hat{\beta}_1 \hat{x}_t + \hat{\beta}_2 x_{t-1} + \hat{\beta}_3 x_{t-2} \\ \hat{x}_{t+2} &= \hat{\beta}_1 \hat{x}_{t+1} + \hat{\beta}_2 \hat{x}_t + \hat{\beta}_3 x_{t-1} \\ \hat{x}_{t+3} &= \hat{\beta}_1 \hat{x}_{t+2} + \hat{\beta}_2 \hat{x}_{t+1} + \hat{\beta}_3 \hat{x}_t\end{aligned}$$

Of course, the quality of the prediction decreases with the considered time horizon.

Autoregression

Example (cont'd) : 5 days temperature forecast from the 26/4/23 to 31/12/24.



Autoregression

How do we choose the optimal number of lags? As any model, the quality of an $AR(p)$ may be measured by its log-likelihood, $\ln L(p)$. If we have T observations and an $AR(p)$ model,

$$L(p) := \prod_{t=p+1}^T f(x_t | \hat{\beta}) ,$$

where $f(\hat{x}_t | \hat{\beta})$ is the pdf of a normal distribution $\mathcal{N}(\hat{x}_t, \hat{\sigma}^2)$

$$f(\hat{x}_t | \hat{\beta}) = \frac{1}{\hat{\sigma} \sqrt{2\pi}} \exp \left(-\frac{1}{2} \frac{(x_t - \hat{x}_t)^2}{\hat{\sigma}^2} \right) .$$

We should then always choose the model with the highest log-likelihood... but , it doesn't account for the **complexity** of the model.

Information Criterion

We may imagine a regression with as many parameters as observations, the fit would be perfect but the predictive power is null (**overfitting**). This is where Information Criterion comes in.

The **Akaike information criterion (AIC)** is a metric that balances model fit and complexity. It is calculated as:

$$AIC = 2p - 2 \ln(L(p))$$

where:

- ▶ p is the number of parameters in the model,
- ▶ $L(p)$ is the maximum value of the likelihood function for the model.

The AIC penalizes models with more parameters to prevent overfitting. **A lower AIC value indicates a better model** because it suggests a good fit with fewer parameters.

Information Criterion

There exists a second metrics:

The **Bayesian information criterion (BIC)** is a metric that balances model fit and complexity. It is calculated as:

$$BIC = \ln(n)p - 2 \ln(L(p))$$

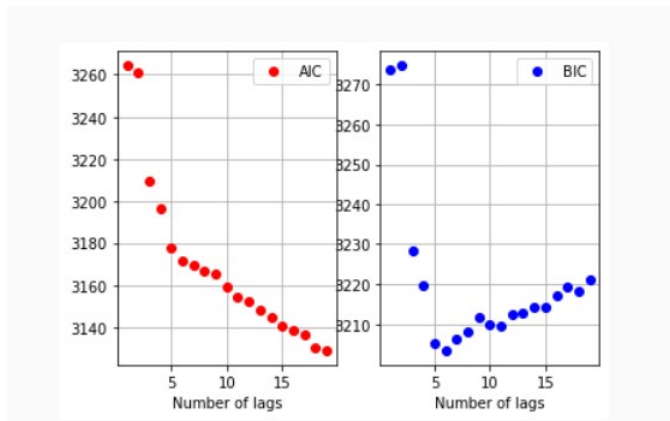
where:

- ▶ n is the number of data
- ▶ p is the number of parameters in the model,
- ▶ $L(p)$ is the maximum value of the likelihood function for the model.

BIC imposes a stronger penalty for additional parameters compared to AIC. A model with a low BIC is preferable. Compared to AIC, BIC tends to favor more simple model.

Information Criterion

Example (cont'd) : According to the BIC, a model with 6 lags is optimal.



Remark : in practise we also use partial autocorrelations but this is out of the scope of this course.

Gaussian Process Regression (GPR)

GP Regression

Compared to linear regression, what are the advantages of a GPR?

- ▶ Non-linearity and Flexibility
 - ▶ Linear Regression assumes a fixed linear relationship between input features and the output.
 - ▶ GPR is a non-parametric model that can capture complex, non-linear relationships without explicitly specifying the form of the function.
- ▶ Uncertainty Quantification
 - ▶ Linear Regression provides confidence intervals under certain assumptions (e.g., homoscedasticity, normality of errors), but these are often limited.
 - ▶ GPR naturally provides a full predictive distribution (mean and variance) for each prediction, offering a principled way to quantify uncertainty.

GP Regression

We want to regress n realizations y_i of Y_i on d explanatory factors $\mathbf{x}_i^\top = (x_{i,1}, \dots, x_{i,d})^\top \in \mathcal{X}$, a subset of $\mathbb{R}^d$.

A **Gaussian process (GP)** $\{f(\mathbf{x}), \mathbf{x} \in \mathcal{X}\}$ is such that for any $n \in \mathbb{N}$ and $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathcal{X}$, $(f(\mathbf{x}_1), \dots, f(\mathbf{x}_n))$ has a joint multivariate Gaussian distribution with a mean function

$$\mathbb{E}(f(\mathbf{x})) = m(\mathbf{x}) \quad \forall \mathbf{x} \in \mathcal{X}$$

and a covariance matrix is defined by a **kernel** $k(\mathbf{x}, \mathbf{x}')$

$$\mathbb{C}(f(\mathbf{x}), f(\mathbf{x}')) = k(\mathbf{x}, \mathbf{x}') \quad \forall (\mathbf{x}, \mathbf{x}') \in \mathcal{X} \times \mathcal{X}.$$

The covariance $n \times n$ matrix is

$$k(\mathbf{X}, \mathbf{X}) = (k(\mathbf{x}_i, \mathbf{x}_j))_{i,j=1,\dots,n}.$$

Key property: For any finite subset of inputs $\{\mathbf{x}_1, \dots, \mathbf{x}_n\} \subset \mathcal{X}$, the corresponding function values

$$(f(\mathbf{x}_1), \dots, f(\mathbf{x}_n)) \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$$

where

- ▶ $\boldsymbol{\mu} = (m(\mathbf{x}_1), \dots, m(\mathbf{x}_n))^{\top}$
- ▶ $\boldsymbol{\Sigma} = k(\mathbf{X}, \mathbf{X})$.

Typically, we assume $m(\mathbf{x}) = 0$ unless prior information about the mean is available. The kernel is chosen a-priori.

GP Regression

GPR approximates a function whose functional form is unknown but whose responses $\mathbf{y} = (y_i)_{i=1,\dots,n}$ are observed for $\mathbf{X} = \{\mathbf{x}_i\}_{i=1,\dots,n}$.

We postulate that responses y_i 's are related to the $\mathbf{x}_i$'s by a GP $f(\cdot)$:

$$y_i = f(\mathbf{x}_i) + \epsilon_i$$

where $\epsilon_i \sim \mathcal{N}(0, \sigma_\epsilon^2)$ is a noise. With $m(\mathbf{x}) = 0$, $\mathbf{y}$ is the realization of $\mathbf{Y}$ a multivariate normal r.v. :

$$\mathbf{Y} \sim \mathcal{N}(0, k(\mathbf{X}, \mathbf{X}) + \sigma_\epsilon^2 I_n)$$

As this is based on a-priori assumptions, this is called the **prior distribution** of responses.

GP Regression

Let us now consider a vector of features $\mathbf{x}_*$, for which the response y_* is not observed, i.e. $\mathbf{x}_* \notin \mathbf{X}$. The joint-distribution of the corresponding $f(\mathbf{x}_*)$ and $\mathbf{Y}$ is:

$$\begin{pmatrix} f(\mathbf{x}_*) \\ \mathbf{Y} \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{bmatrix} k(\mathbf{x}_*, \mathbf{x}_*) & k(\mathbf{X}, \mathbf{x}_*) \\ k(\mathbf{X}, \mathbf{x}_*)^\top & k(\mathbf{X}, \mathbf{X}) + \sigma_\epsilon^2 I_n \end{bmatrix} \right)$$

If we observe realization $\mathbf{y}$ of $\mathbf{Y}$, the estimator $\hat{y}_*$ of y_* is the following conditional expectation:

$$\hat{y}_* = \mathbb{E}(f(\mathbf{x}_*) | \mathbf{Y} = \mathbf{y}) .$$

But the distribution of $f(\mathbf{x}_*) | \mathbf{Y} = \mathbf{y}$ is known! (See the conditional distribution of a multivariate normal.

Proposition From the property of the multivariate normal, the **posterior** distribution of $f(\mathbf{x}_*) \mid \mathbf{Y} = \mathbf{y}$ is normal $\mathcal{N}(\hat{\mathbf{y}}_*, \sigma_*^2)$ with a mean

$$\begin{aligned}\hat{\mathbf{y}}_* &= \mathbb{E}(f(\mathbf{x}_*) \mid \mathbf{Y} = \mathbf{y}) \\ &= k(\mathbf{x}_*, \mathbf{X})^\top (k(\mathbf{X}, \mathbf{X}) + \sigma_\epsilon^2 I_n)^{-1} \mathbf{y}\end{aligned}$$

and a variance equal to

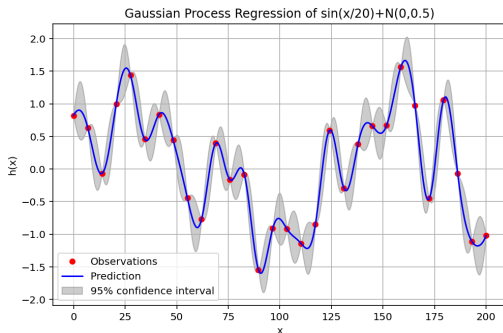
$$\sigma_*^2 = k(\mathbf{x}_*, \mathbf{x}_*) - k(\mathbf{x}_*, \mathbf{X})^\top (k(\mathbf{X}, \mathbf{X}) + \sigma_\epsilon^2 I_n)^{-1} k(\mathbf{X}, \mathbf{x}_*).$$

σ_ϵ^2 is a Tikhonov regularization parameter that allows to invert ill-conditioned matrix $k(\mathbf{X}, \mathbf{X})$. As it is a parameter a priori chosen, you can set it to zero if the matrix is well conditioned.

GP Regression

Example: GPR of a function, $f(x) = \sin(x/20) + \epsilon$ and $\epsilon \sim N(0, 0.5)$ with a kernel

$$k(\mathbf{x}, \mathbf{x}') = 0.814^2 \exp((x - x')^2 / (2 \times 5.09^2)) .$$



The covariance matrix in a GP context, is determined by the kernel function $k(x, x')$, which encodes the similarity between inputs.

A valid kernel $k(x, x')$ satisfies the following properties:

- ▶ Symmetry: $k(x, x') = k(x', x)$,
- ▶ Positive semi-definiteness: $\mathbf{u}^\top k(\mathbf{X}, \mathbf{X}) \mathbf{u} \geq 0$, $\forall \mathbf{u} \in \mathbb{R}^n$.

Role in GPs: The kernel $k(x, x')$ governs the smoothness, periodicity, and other properties of functions modeled by the GP.

There exists several kernel functions but a popular one is the radial basis function.

Radial Basis Function (RBF) Kernel:

$$k(x, x') = \eta^2 \exp\left(-\frac{\|x - x'\|^2}{2h^2}\right),$$

where h is the bandwidth and η is a dispersion parameter.

- ▶ The RBF kernel assumes that points closer in the input space are more strongly correlated.
- ▶ This kernel approximates smooth functions in the GPR.

Matern and RQ Kernels

If we denote by $d = ||x - x'||^2$, the **Matern kernel** is defined as:

$$k_{\nu}(d) = \sigma^2 \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\sqrt{2\nu} \frac{d}{\rho} \right)^{\nu} K_{\nu} \left(\sqrt{2\nu} \frac{d}{\rho} \right),$$

where Γ is the gamma function, K_{ν} the Bessel function of second order, ρ and ν are strictly positive parameters.

The **Rational Quadratic** kernel is defined as:

$$k(d) = \left(1 + \frac{d^2}{2\alpha l^2} \right)^{-\alpha},$$

where α is the scale mixture parameter, l is the length scale of the kernel.

Hyper-parameters

Given that $\mathbf{Y}$ is a-priori normal

$$\mathbf{Y} \sim \mathcal{N}(0, k(\mathbf{X}, \mathbf{X}) + \sigma_\epsilon^2 I_n)$$

the kernel hyper-parameters, (h, η) are found by log-likelihood maximization

$$\log l(h, \eta) \propto -\frac{1}{2} \log |k(\mathbf{X}, \mathbf{X}) + \sigma_\epsilon^2 I_n| - \frac{1}{2} \mathbf{y}^\top (k(\mathbf{X}, \mathbf{X}) + \sigma_\epsilon^2 I_n)^{-1} \mathbf{y}$$

This is implemented in scikit learn

```
from sklearn.gaussian_process import GaussianProcessRegressor
from sklearn.gaussian_process.kernels import RBF
kernel = RBF()
gp = GaussianProcessRegressor(kernel=kernel,
                              n_restarts_optimizer=10)

# Fit to the data
gp.fit(x, y)
```